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SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Department of Biomedical Engineering

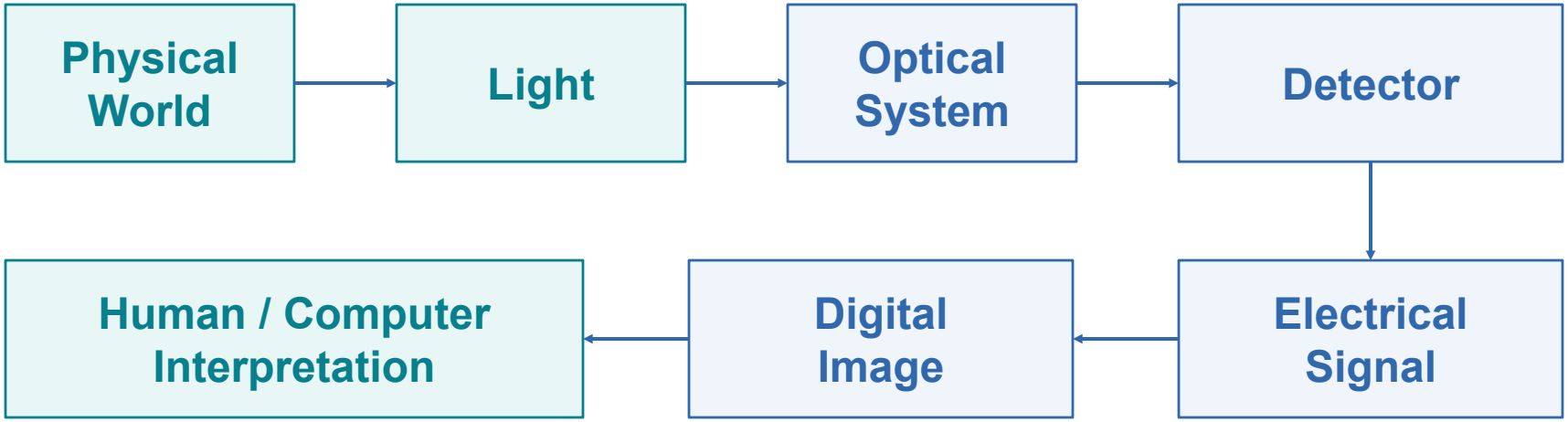
Digital Image Fundamentals

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Course Roadmap

Image formation is a chain of physical measurements



Part I

How do WE see an image?

Human Visual System & Perception

Part II

How does a MACHINE see an image?

Photodetectors & Image Sensors

Part III

How GOOD is the acquired image?

Noise, QE & SNR

What is measured—and what can be lost—at each step?

Elements of Visual Perception

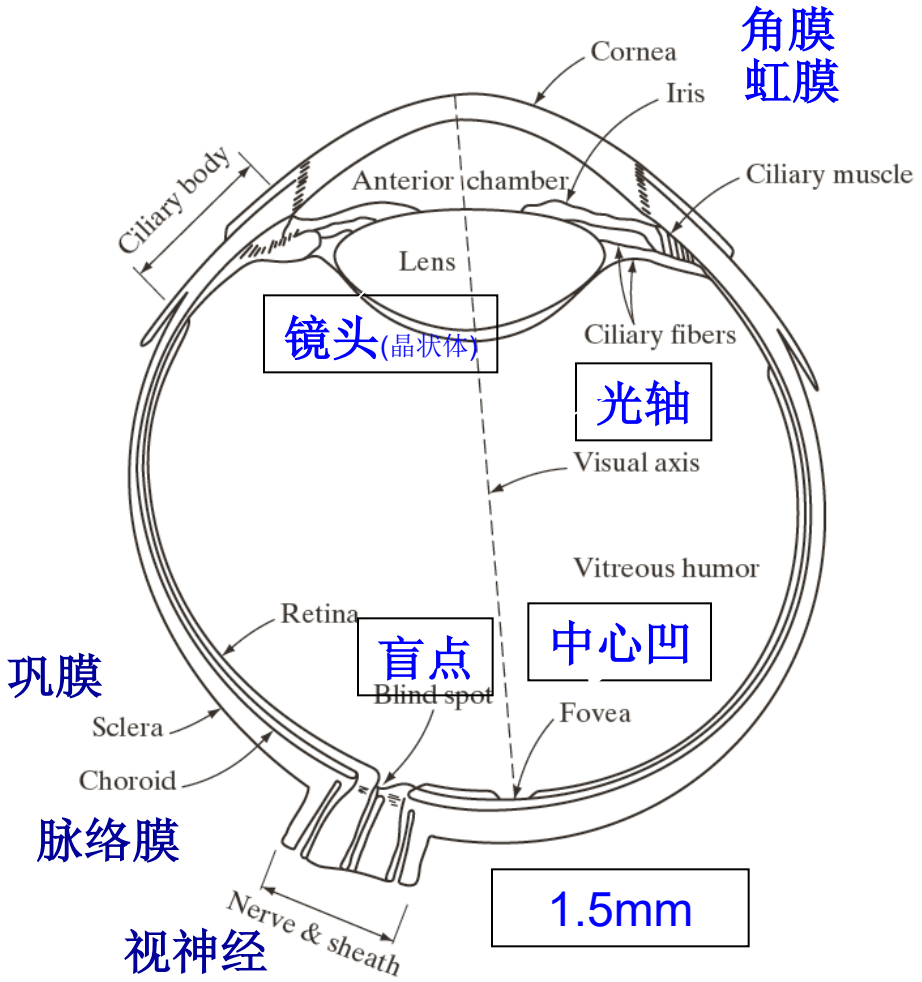


Why study human vision?

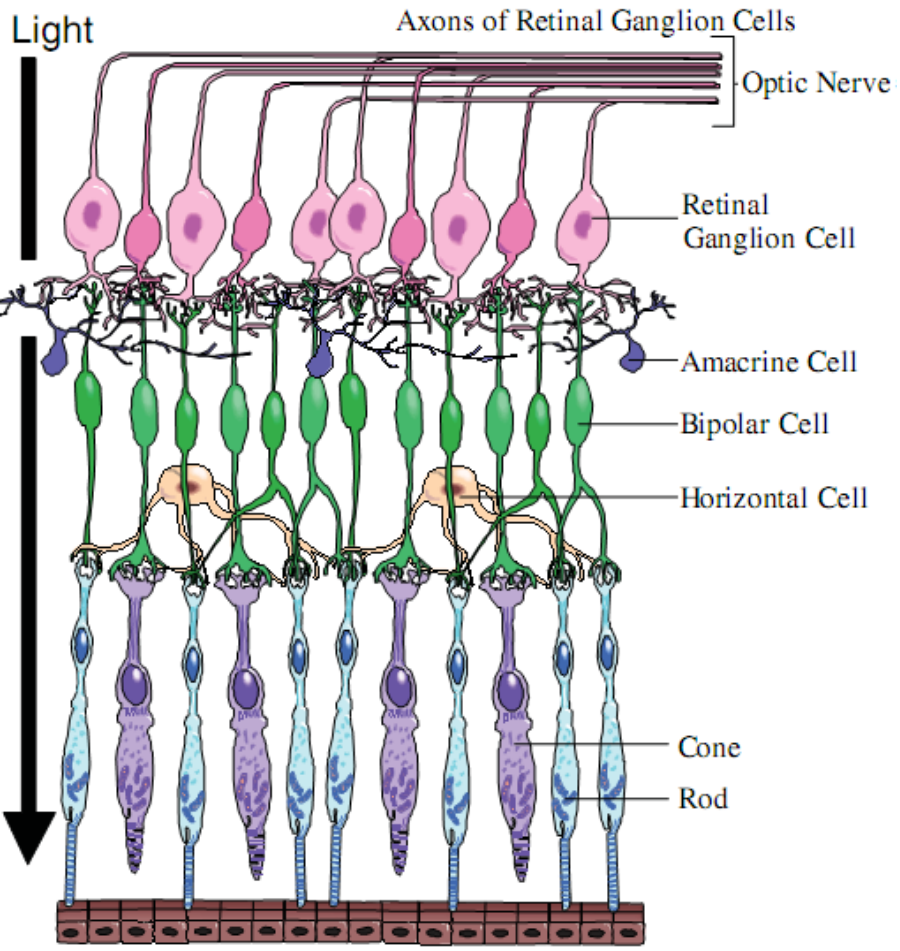
- The quality evaluation of digital image processing is mostly subjectively evaluated by the human eye.
- Many image processing related technologies should be designed according to the characteristics of the human eye: such as screen refresh frequency, image quantization level, stereo display (binocular vision), image compression
- According to the mechanism of human cognition of image content, new image processing algorithms can be researched

Structure of human eye

- Approximately spherical, with an average diameter of about 20mm, the eyeball wall is composed of three layers of membranes:
  1. Cornea and sclera
  2. Choroid, ciliary body, iris
  3. Retina
- The inside of the eyeball is mainly the lens and vitreous
- The eyeball is divided into two parts: the refractive system and the photosensitive system



Structure of human eye

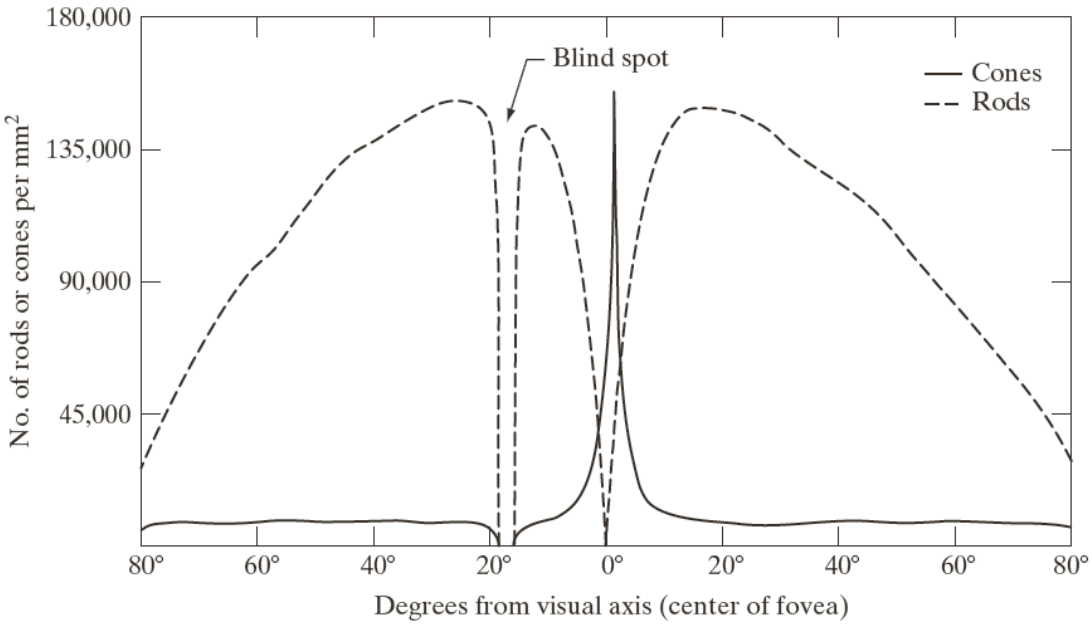


- **Photosensitive system** converts light stimulation into nerve stimulation and send it to the brain nervous system. It is mainly composed of the retina.
- **Retina** is an inverted structure, composed of three layers of cells from the outside to the inside.

1. **Ganglion cell layer**: Ganglion cells, which are connected to the optic nerve and transmit nerve stimulation
2. **Bipolar cell layer**: Bipolar cells, connecting Photoreceptor cells and ganglion cells.
3. **Photoreceptor cell layer**: two types of visual cells, cone-shaped cells and rod-shaped cells, which feel the light intensity and color stimulation

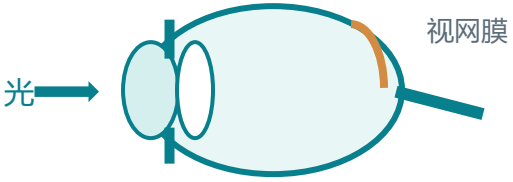
Structure of human eye

- Cone cells
  - Number: 6~7 million, located near fovea
  - Each cone cell is connected to a single bipolar cell
  - High spatial resolution, high sensitivity to color
  - Work only in bight light (photopic or bright-light vision)
- Rod cells
  - Number: 75 million to 150 million
  - Several rod cells connected to the same bipolar cell
  - Low spatial resolution, no color perception
  - Night vision (scotopic or dim-light vision)



Human Eye vs. Camera

HUMAN EYE 人眼



DIGITAL CAMERA 数字相机



Cornea + lens 角膜 + 晶状体	聚焦	**Camera lens** 相机镜头
Iris / Pupil 虹膜 / 瞳孔	控光	**Aperture** 光圈 / 通光孔径
Retina 视网膜	成像面	**Image sensor** 图像传感器
Rod / Cone 视杆 / 视锥细胞	感光	**Photodetector / Pixel** 光电探测单元 / 像素
Optic nerve 视神经	传输	**Data transmission / readout** 数据传输 / 读出
Brain 大脑	处理	**Image processing / interpretation** 图像处理 / 解释

功能类比，并非一一等同；视网膜本身也参与早期信息处理。

The human eye is more like a camera + image processor than a camera alone.

Image Formation in the Eye

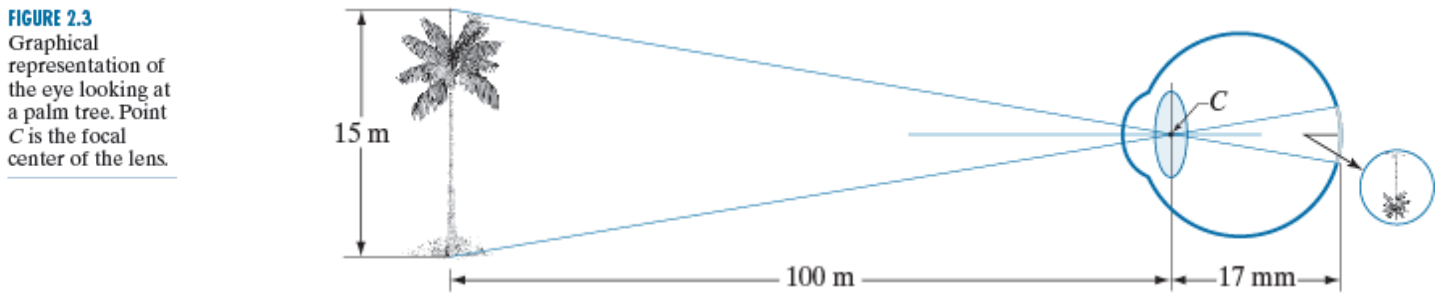


FIGURE 2.3
Graphical representation of the eye looking at a palm tree. Point *C* is the focal center of the lens.

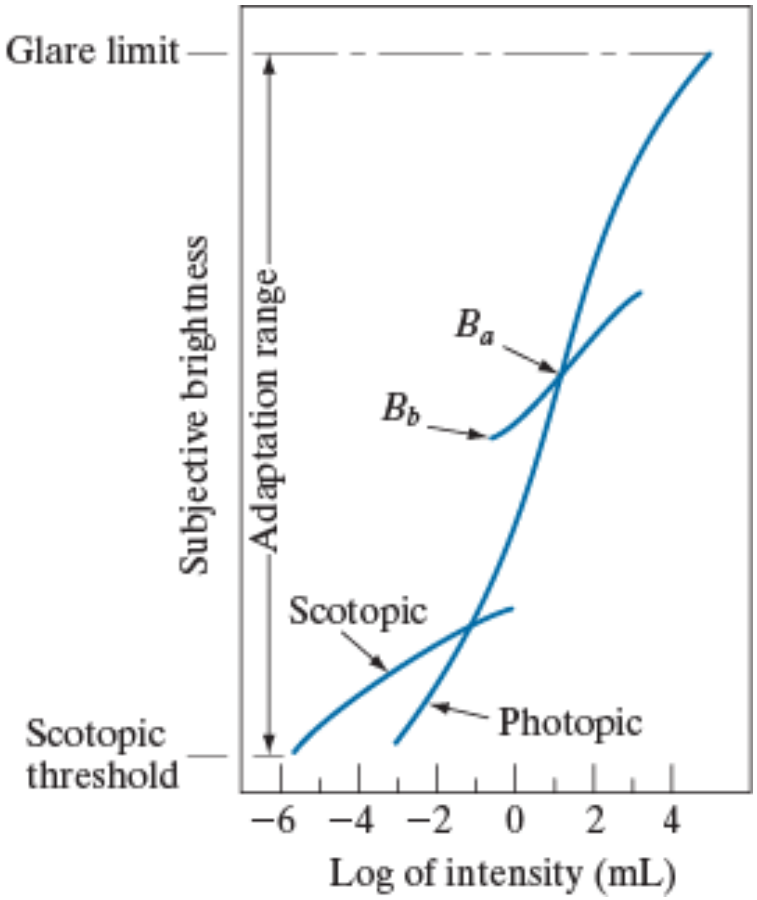
Focusing in a Camera vs. the Human Eye

- **Camera:** The focal length of the lens is typically fixed. Objects at different distances are brought into focus by changing the **distance between the lens and the image sensor**.
- **Human eye:** The distance between the **lens and the retina remains essentially fixed**. Instead, the eye focuses on objects at different distances by **changing the shape—and therefore the focal length—of the lens**, a process known as **accommodation**.

Image Formation in the Eye

- Brightness adaptation and discrimination:

The human eye can adapt to a wide range of light intensity levels, on the order of 10^{10}, Much higher than artificial imaging and display devices

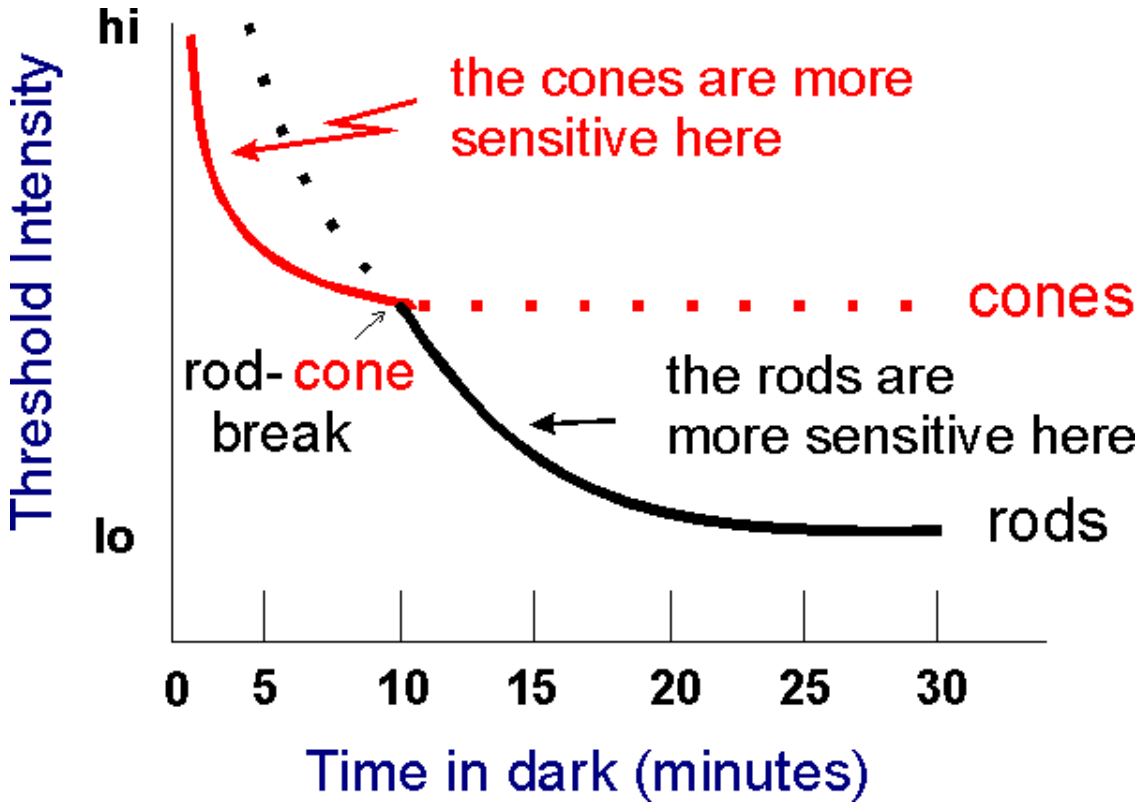


$$b = \log(a)$$

Subjective brightness
主观亮度

Light intensity
光强度

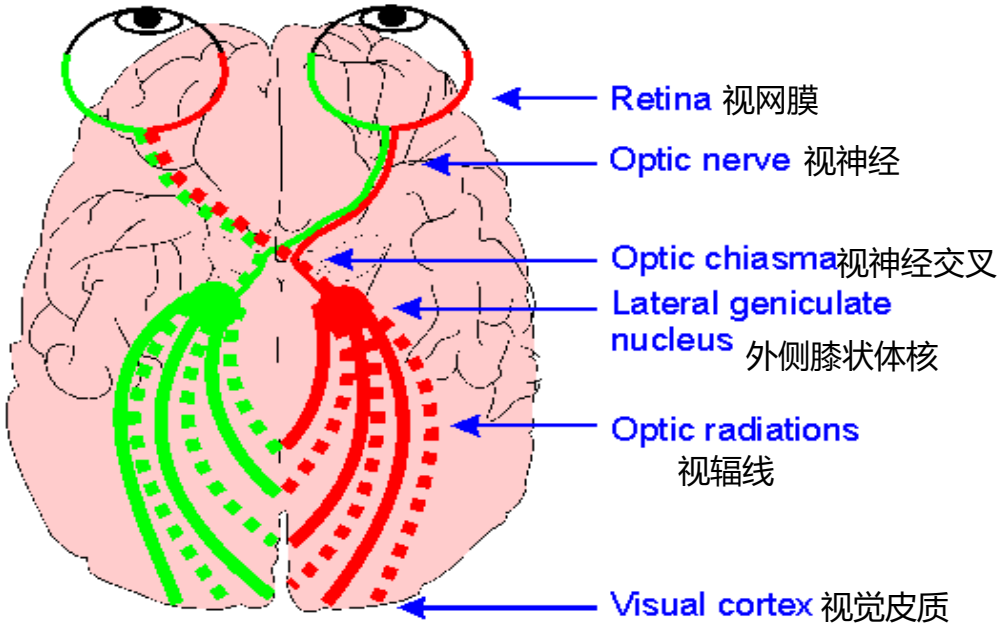
Dark Adaptation Function



For approximately the first 10 minutes in the dark, the cones require less light to reach a threshold response than the rods. Thereafter, the rods require less light.

The point at which the rods become more sensitive is called the rod-cone break.

Neural processing structure



More detailed look at LGN & cortex

Master Diagram

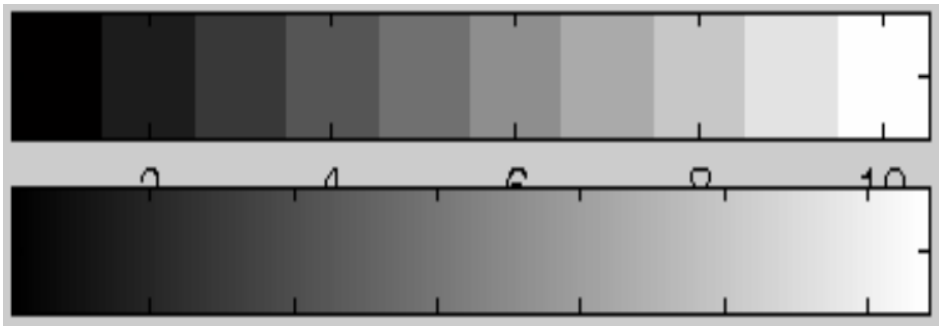
The image formation process in the human eye

The spatial characteristics of human vision

- The spatial resolution is 1’
- The gray resolution is 64 levels

The temporal characteristics of human vision

- When the frame rate of the active image is at least 15 fps, the human eye will have the feeling of continuous image
- When the active image frame rate is 25 fps, the human eye cannot feel the flicker
- Surveillance video 15fps, TV and movie 24fps, computer screen 60fps

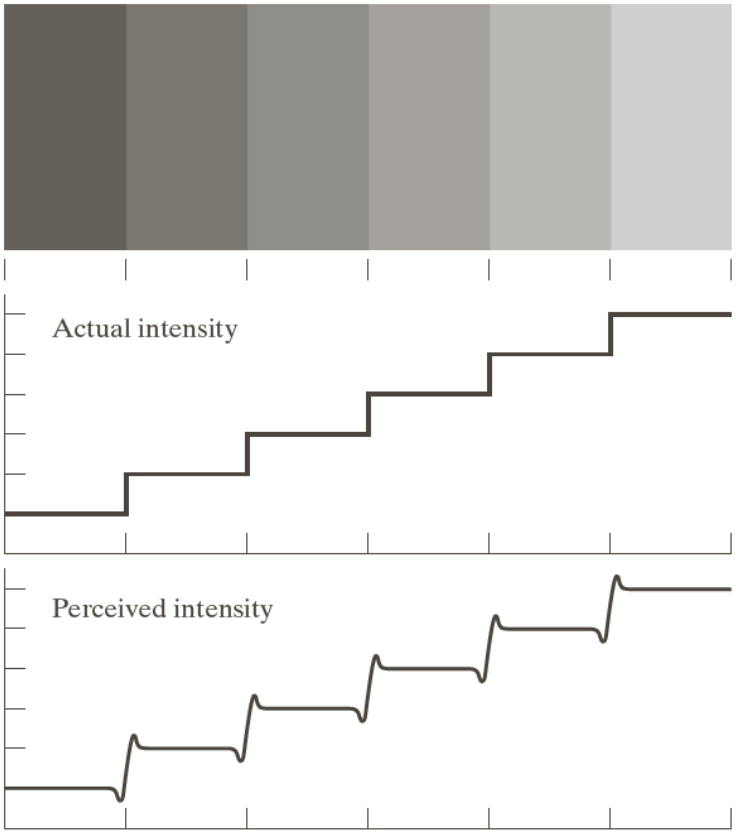


10 gray levels

128 gray levels

Seeing is Believing?

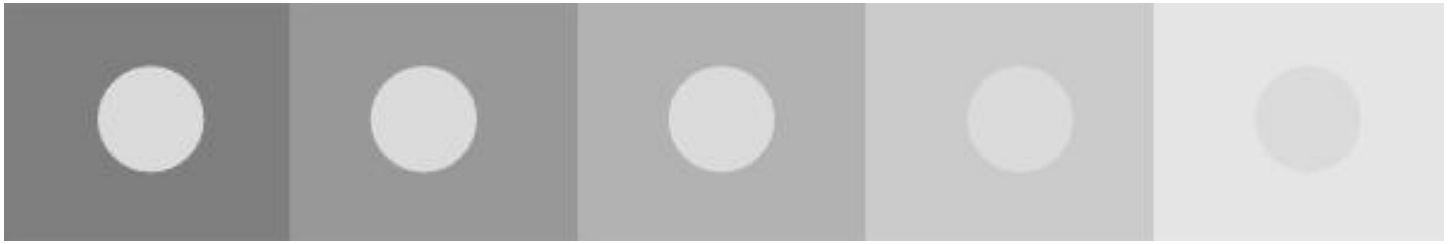
Mach band effect



Perceived brightness is not a simple intensity function; the visual system tends to overestimate or underestimate the boundary value of different brightness areas.

Seeing is Believing?

Simultaneous Contrast

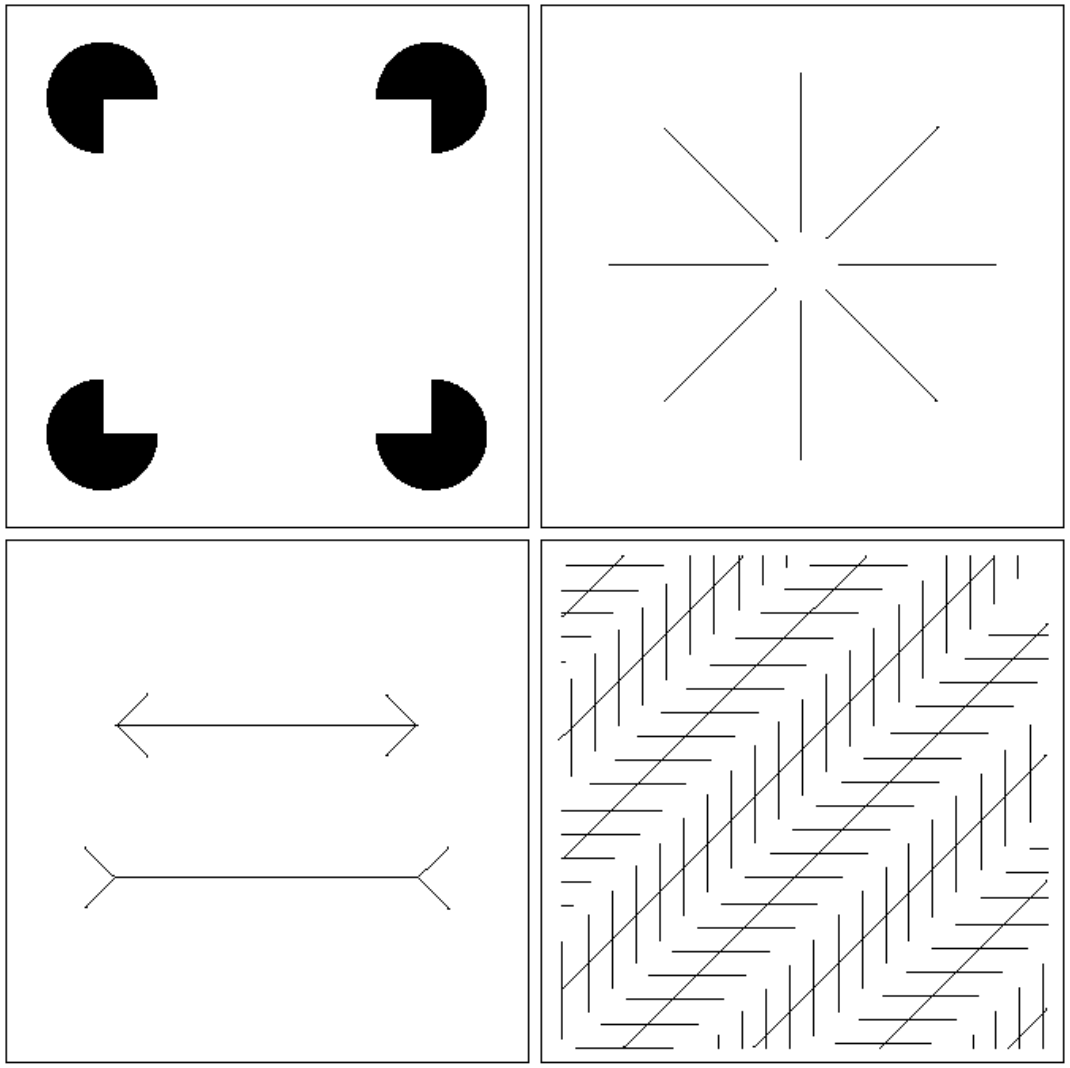


The round dots are all exactly the same color gray. I merely created the one on the extreme right and then copied it to each of the darker rectangles. Clearly, the lightness of the background effects the way in which we perceive the center gray dot. This effect is called simultaneous contrast. Again, as can be seen in other demonstrations, the perception of this kind of display is a result of lateral interactions.

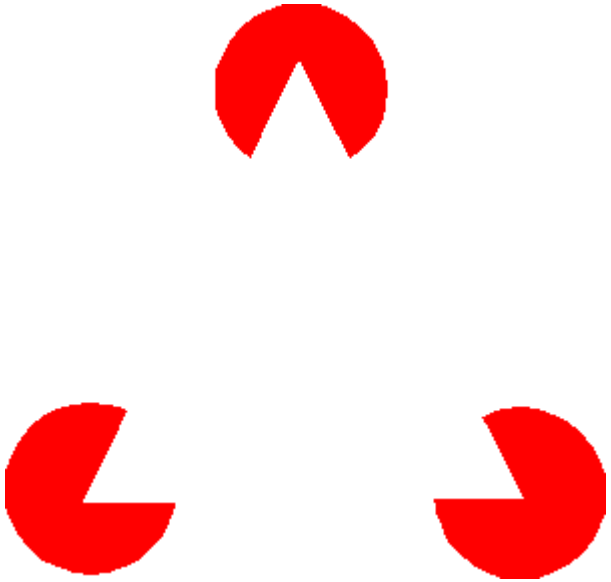
Seeing is Believing?

a b
c d

FIGURE 2.9 Some well-known optical illusions.

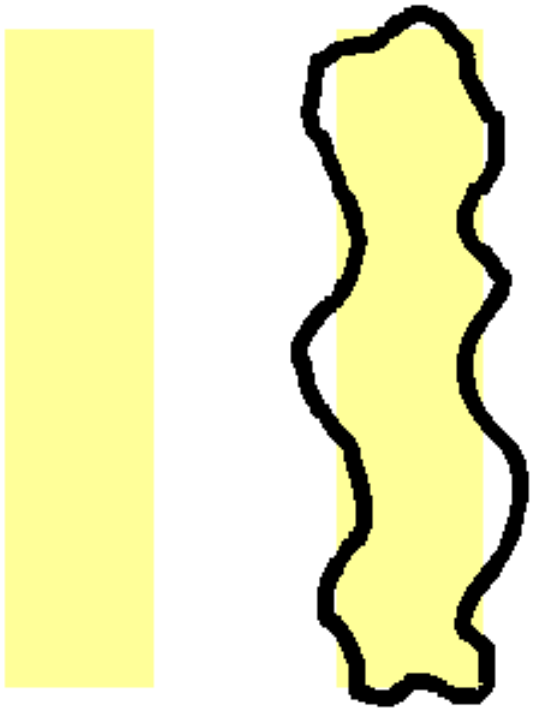


Kanizsa Illusion

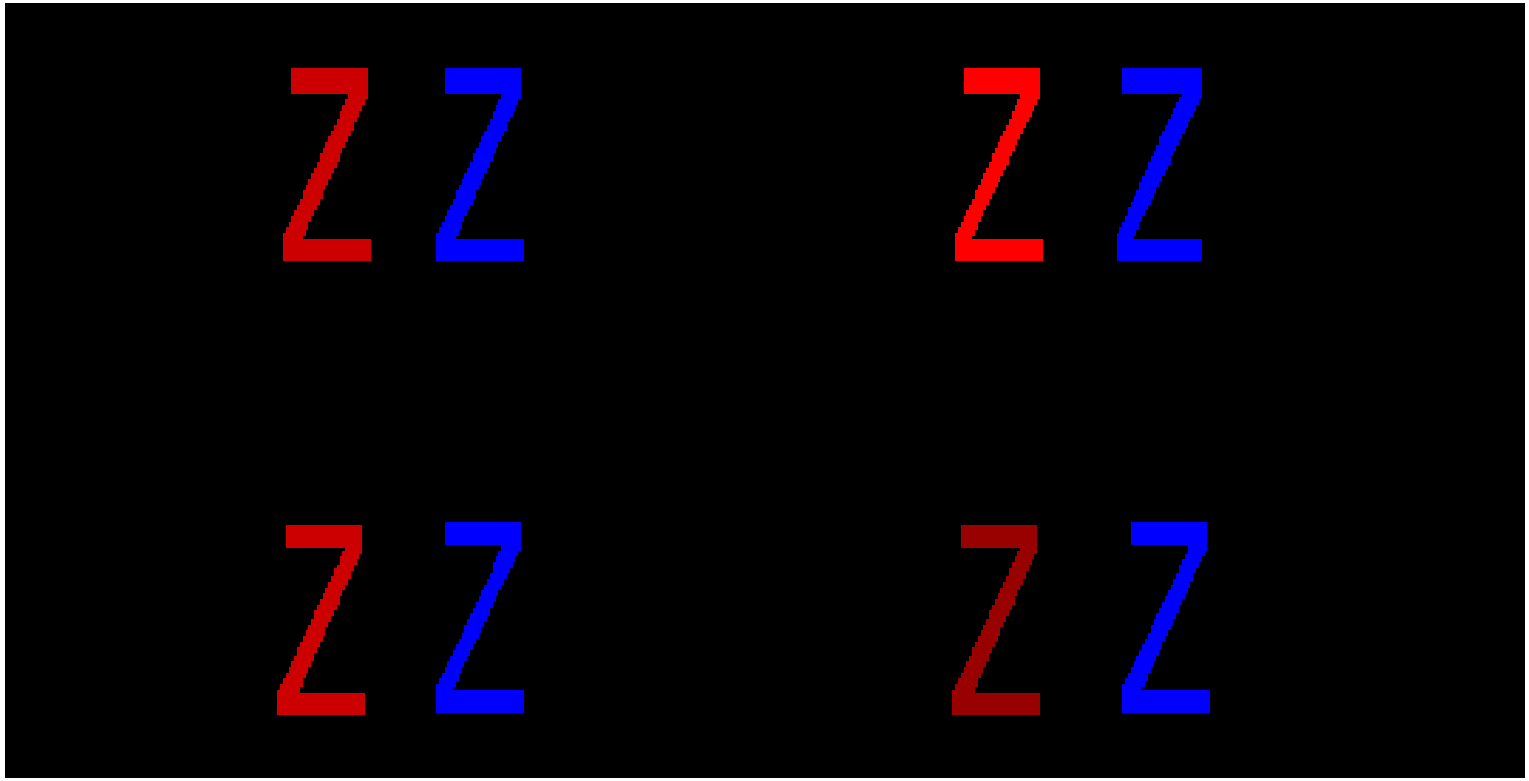


This is called a Kanizsa figure after the person who invented it. If you look carefully you will probably see the edges of the entire triangle, even though the triangle is defined only by the notches in the red disks.

Boynton Illusion



Color Influences Distance Perception

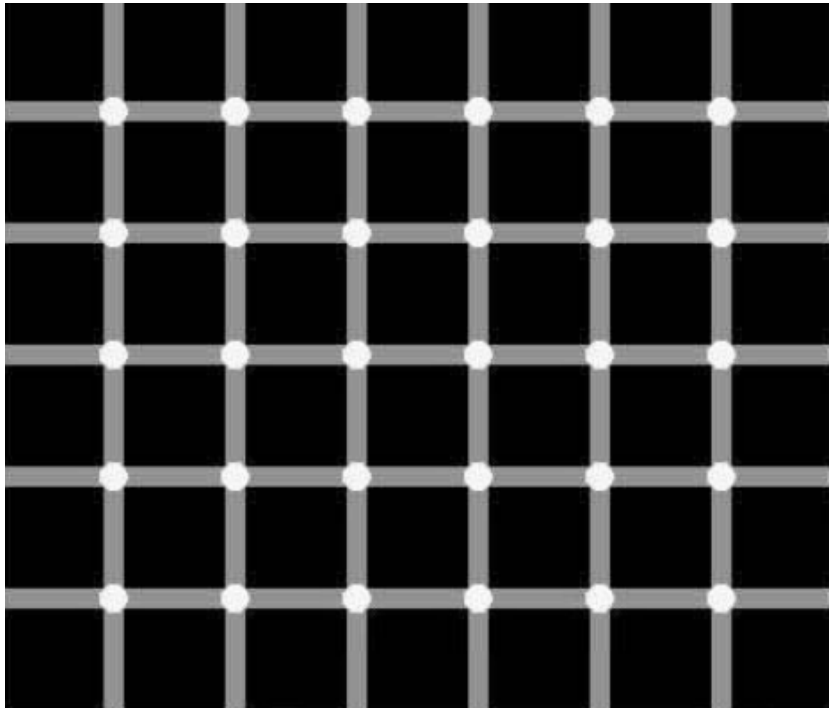


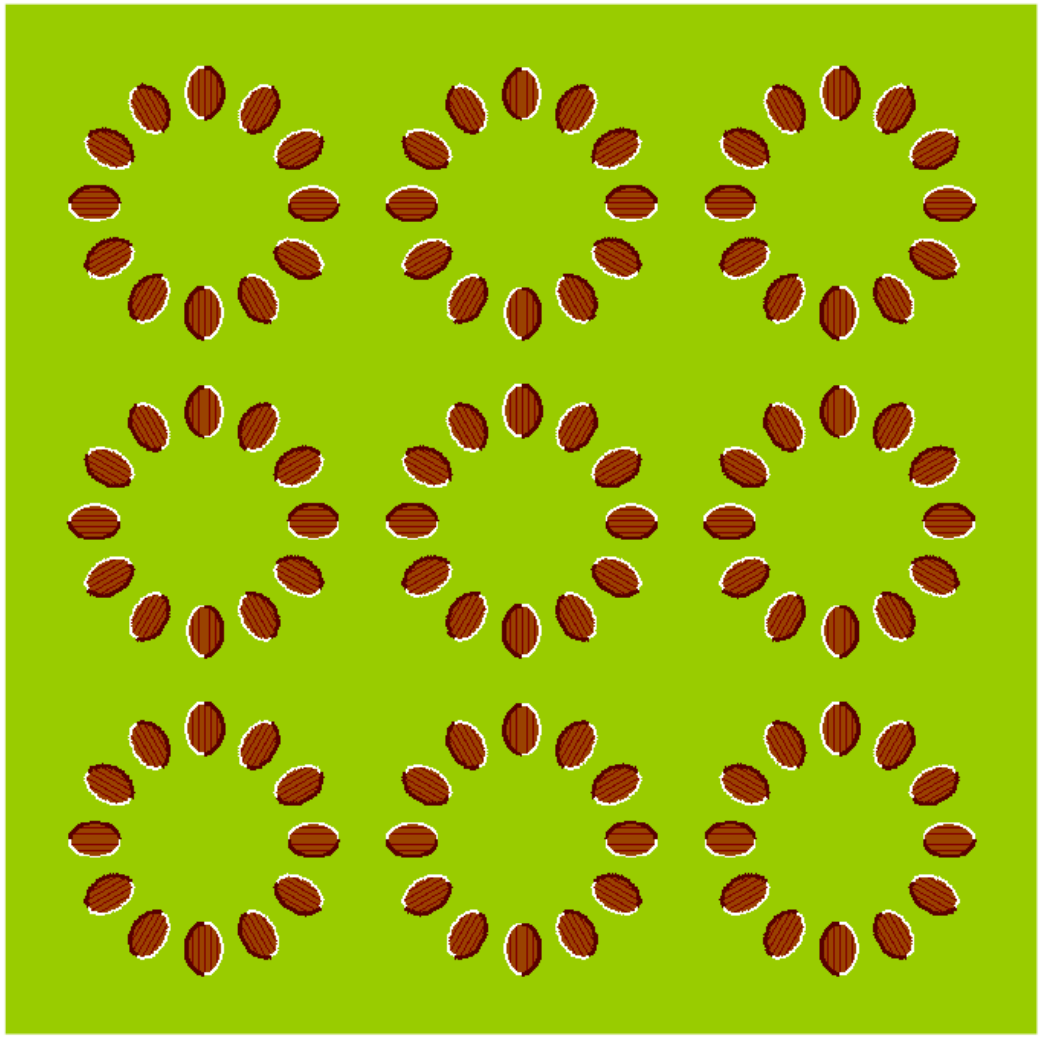
Since these letters are all presented on your video monitor, they obviously are all located in the same plane. But do all the Z's appear to be in the same plane or do some appear closer than others?

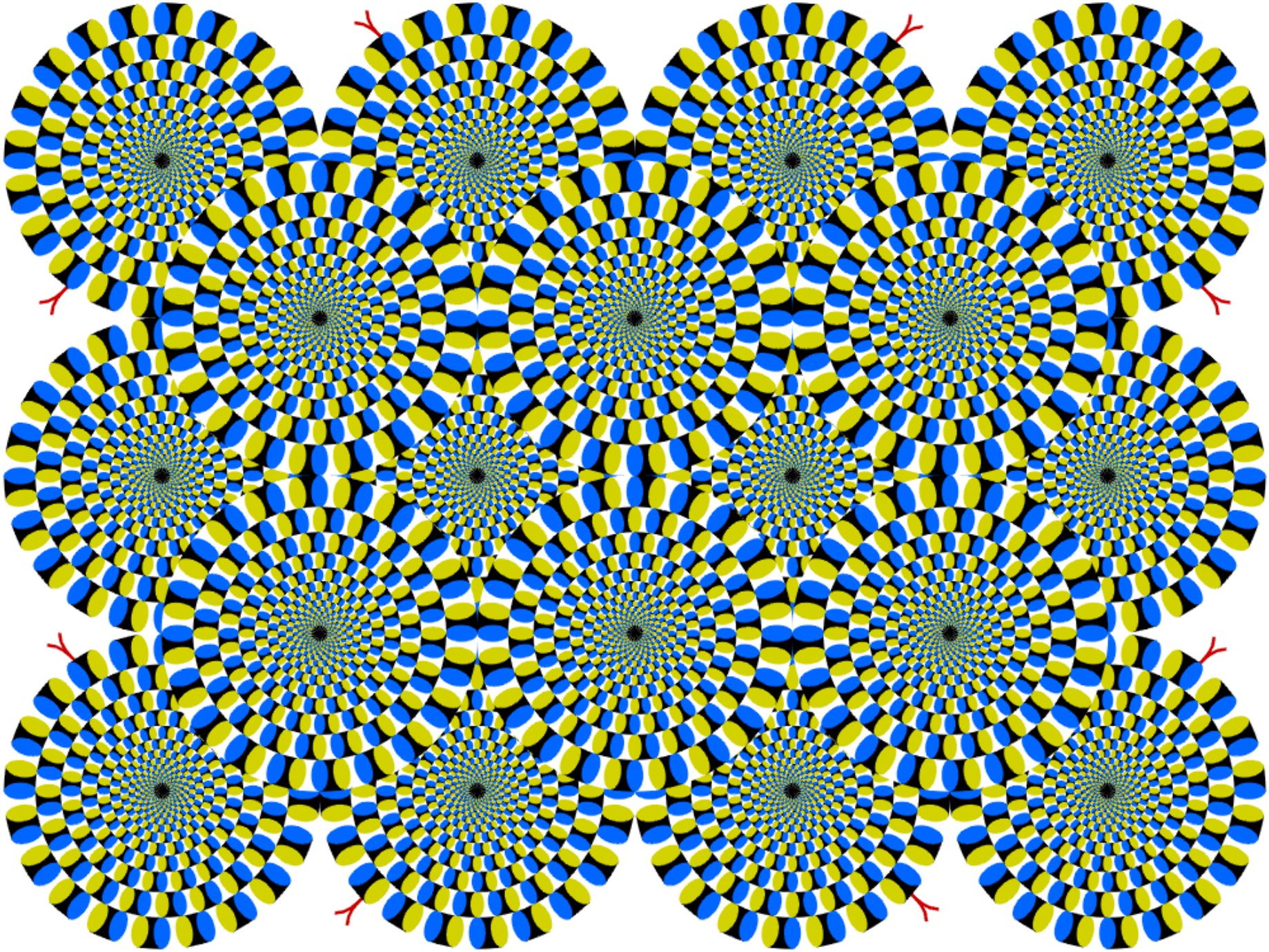
more information on this demonstration

Persistence of vision

- The subjective brightness perception of the human eye is related to the action time of light. Once a light image is formed on the retina, after it disappears, the visual system will feel the light image for a period of time.







From Human Vision to Machine Vision

Human eye



Photoreceptors + retinal processing

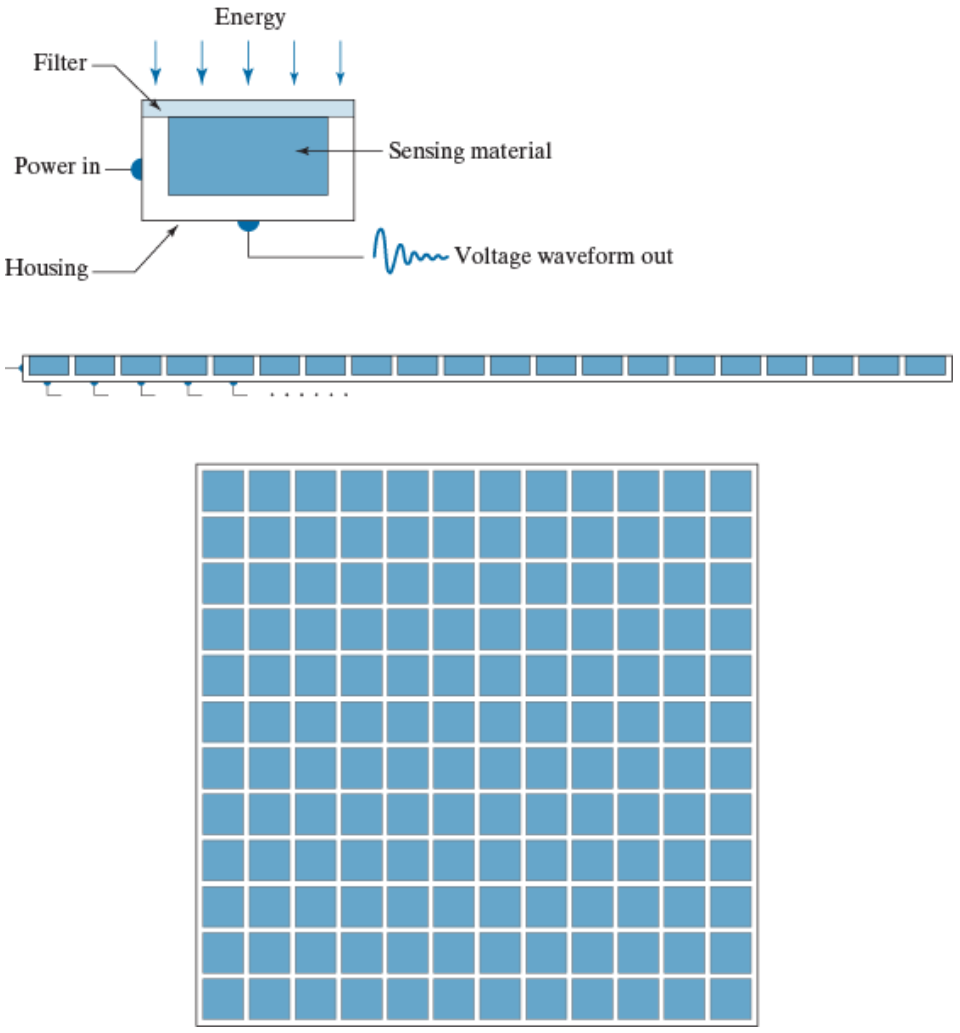
Digital camera



Photodetection + electronic readout

Both systems convert photons into measurable signals.

Electronic Imaging Sensors



a
b
c

FIGURE 2.12
(a) Single sensing element.
(b) Line sensor.
(c) Array sensor.

Image Acquisition Using a Single Sensor

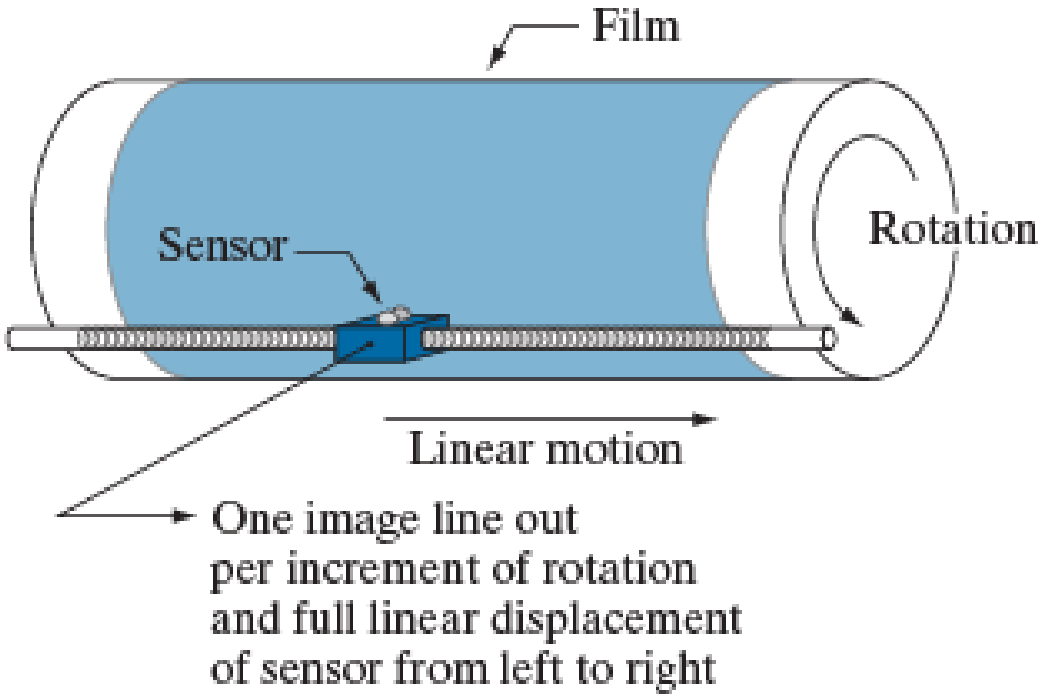
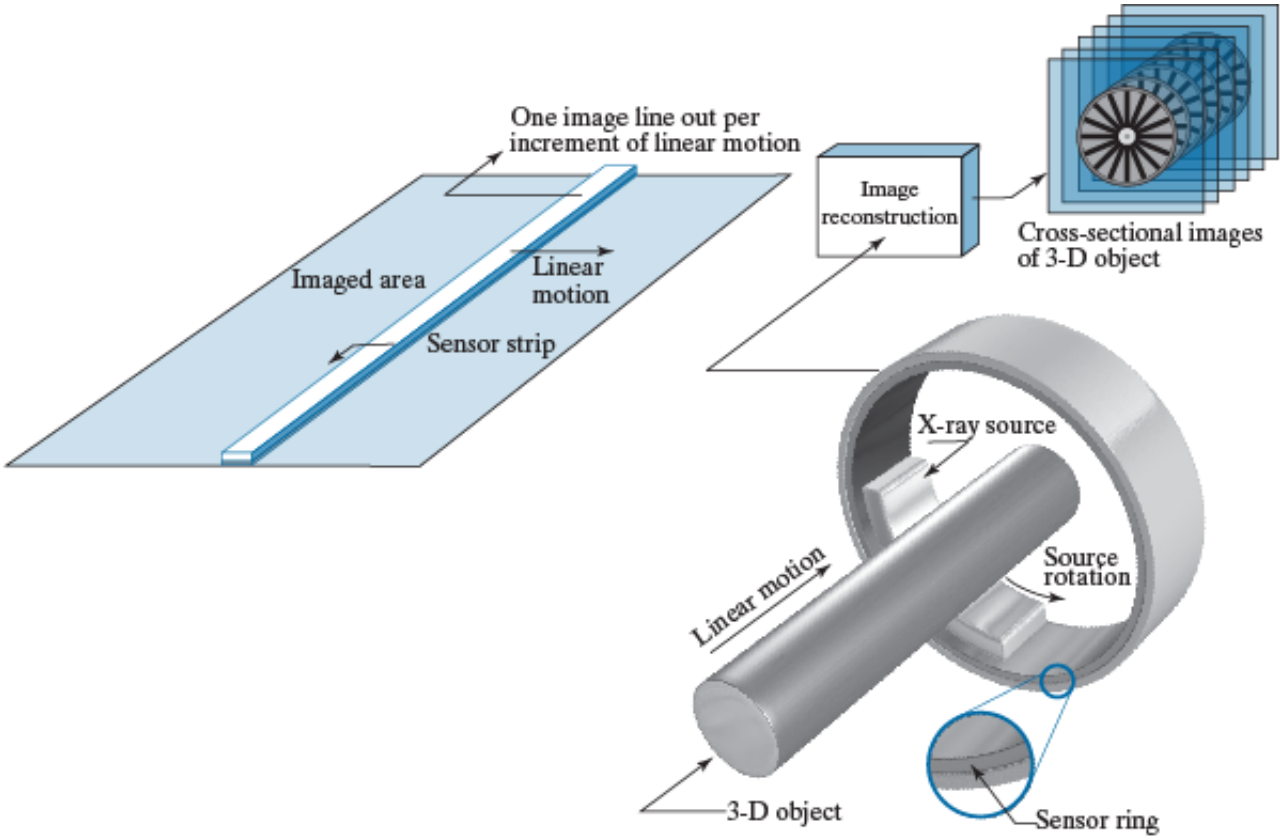


FIGURE 2.13
Combining a single sensing element with mechanical motion to generate a 2-D image.

Image Acquisition Using Sensor Strips

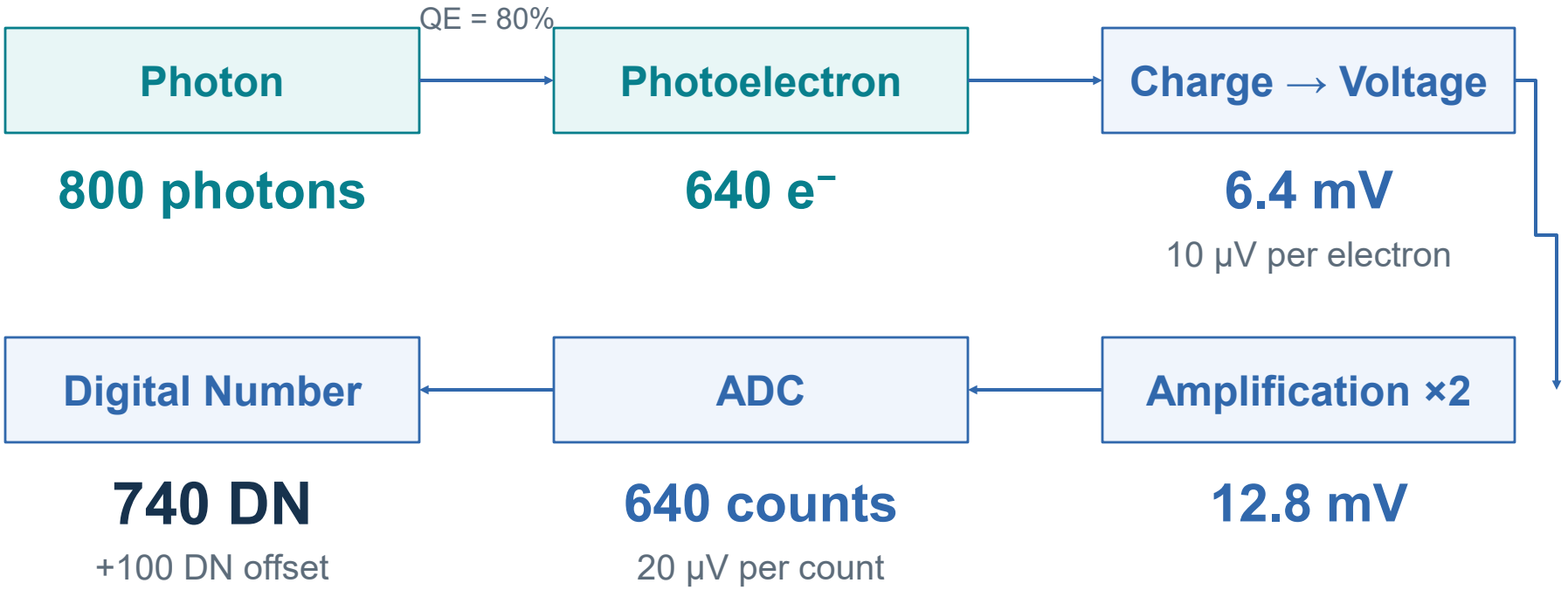


a b

FIGURE 2.14 (a) Image acquisition using a linear sensor strip. (b) Image acquisition using a circular sensor strip.

From Photon to Digital Number

One pixel, one exposure — illustrative mean values



Same photons, gain ×4 instead of ×2 → 1380 DN

Pixel value depends on the acquisition settings.

Photodiode

- **Photodiode is an electronic device that can convert photon energy to electronic energy.** The core of PD is a PN junction. Its major difference with normal diode is that, in order to accept more photons, the size of the junction is as large as possible, and the electrode is as small as possible. The junction gap is usually less than 1um.
- 和普通二极管相比，在结构上不同的是，为了便于接受入射光照，PN结面积尽量做的大一些，电极面积尽量小些，而且PN结的结深很浅，一般小于1微米。
- PD works under inverse voltage. When no incidence present, the inverse current is very small (<0.1uA), it is called dark current.
- 光电二极管是在反向电压作用之下工作的。没有光照时，反向电流很小（一般小于0.1微安），称为暗电流。

PD

- When light illuminate the PD, the photon carrying energy enters the PN junction, and transfers the energy to the bound electrons, so that part of the electrons can break the covalent bond and generate electron-hole pair, called photo-induced carrier.
- 当有光照时，携带能量的光子进入PN结后，把能量传给共价键上的束缚电子，使部分电子挣脱共价键，从而产生电子---空穴对，称为光生载流子。

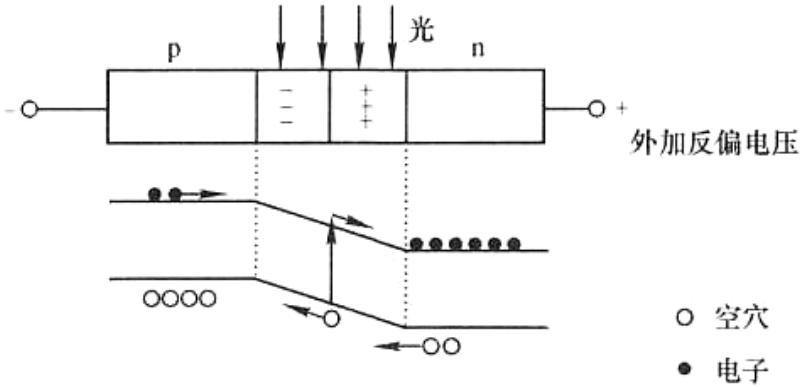


图 1.1.1－1　硅光电二极管工作原理

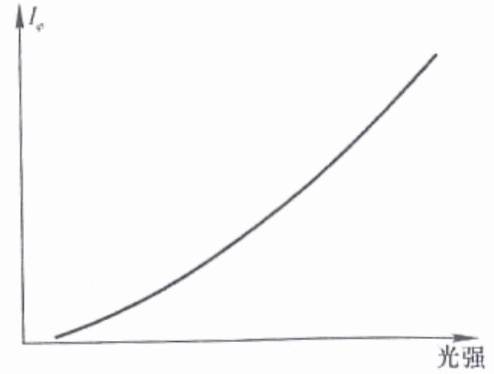


图 1.1.1－4　光电流与入射光强的关系

光电二极管

- The photo-induced carrier joins the drift movement under the inverse voltage, and increases dramatically the inverse current. More intense light generates higher inverse current. This feature is called **photoconduction**. The current generated is called photocurrent. When wiring a load resistor, the voltage will change with the photocurrent.
- 它们在反向电压作用下参加漂移运动，使反向电流明显变大，光的强度越大，反向电流也越大。这种特性称为“光电导”。光电二极管在一般照度的光线照射下，所产生的电流叫光电流。如果在外电路上接上负载，负载上就获得了电信号，而且这个电信号随着光的变化而相应变化。

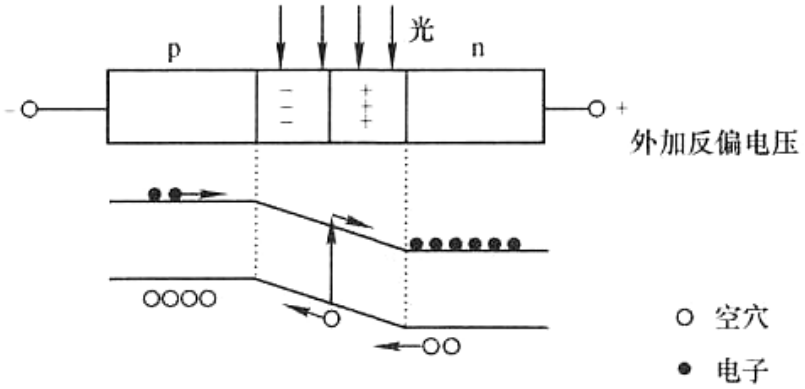


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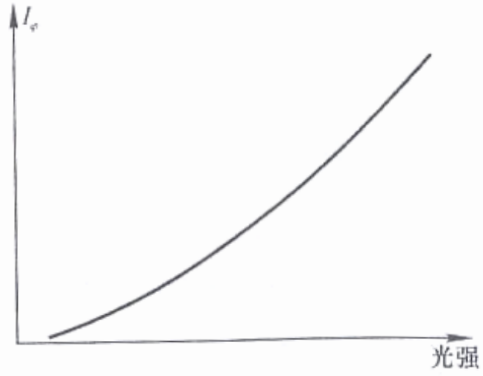


图 1.1.1－4　光电流与入射光强的关系

Q: The photocurrent generated with PD is very small, leading to difficulty in measuring tiny signal such as fluorescence. How can we amplify the signal to make the detection more sensitive?

A: Amplify the signal with one electron generates another, or multiple electrons.

光电倍增管 (Photomultiplier Tube)

- 依据光电子发射、二次电子发射和电子光学的原理制成的、透明真空壳体内装有特殊电极的器件。光阴极在光子作用下发射电子，这些电子被外电场(或磁场)加速，聚焦于第一次极。这些冲击次极的电子能使次极释放更多的电子，它们再被聚焦在第二次极。这样，一般经十次以上倍增，放大倍数可达到10^8～10^{10}。最后，在高电位的阳极收集到放大了的光电流。输出电流和入射光子数成正比。整个过程时间约10^{-8}秒。

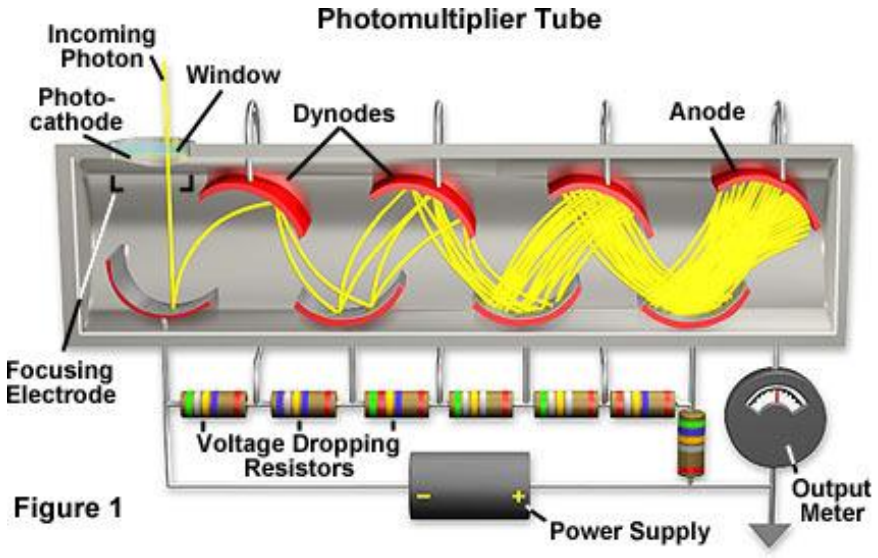


Figure 1

http://www.microscopy.fsu.edu/primer/digitalimaging/concepts/photomultipliers.html

www.bb.ustc.edu.cn/.../student%20works/sw21.html

A photomultiplier tube, useful for light detection of very weak signals, work by amplifying the electrons generated by a photocathode exposed to a photon flux.

Photocathode of PMT

- The spectral response, quantum efficiency, and dark current of a photomultiplier tube are determined by the composition of the photocathode.
- The best photocathodes capable of responding to visible light are less than 30 percent quantum efficient, meaning that 70 percent of the photons impacting on the photocathode do not produce a photoelectron and are therefore not detected.

Photo counting

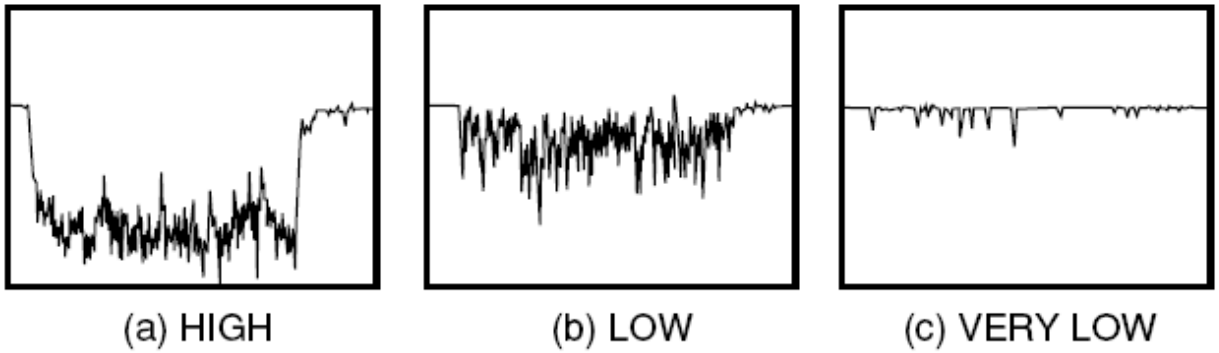


Figure 6-2: Photomultiplier tube output waveforms observed at different light levels

When the incidence is high, PMT detects the average intensity;

When the incidence is very, very low, the signal will be buried in the noise.

When the light is so weak that within one time unit of PMT response, there is at most one photon illuminated, the counted pulse will be the intensity of the weak signal. (There is a threshold between signal and dark current).

Photo-counting

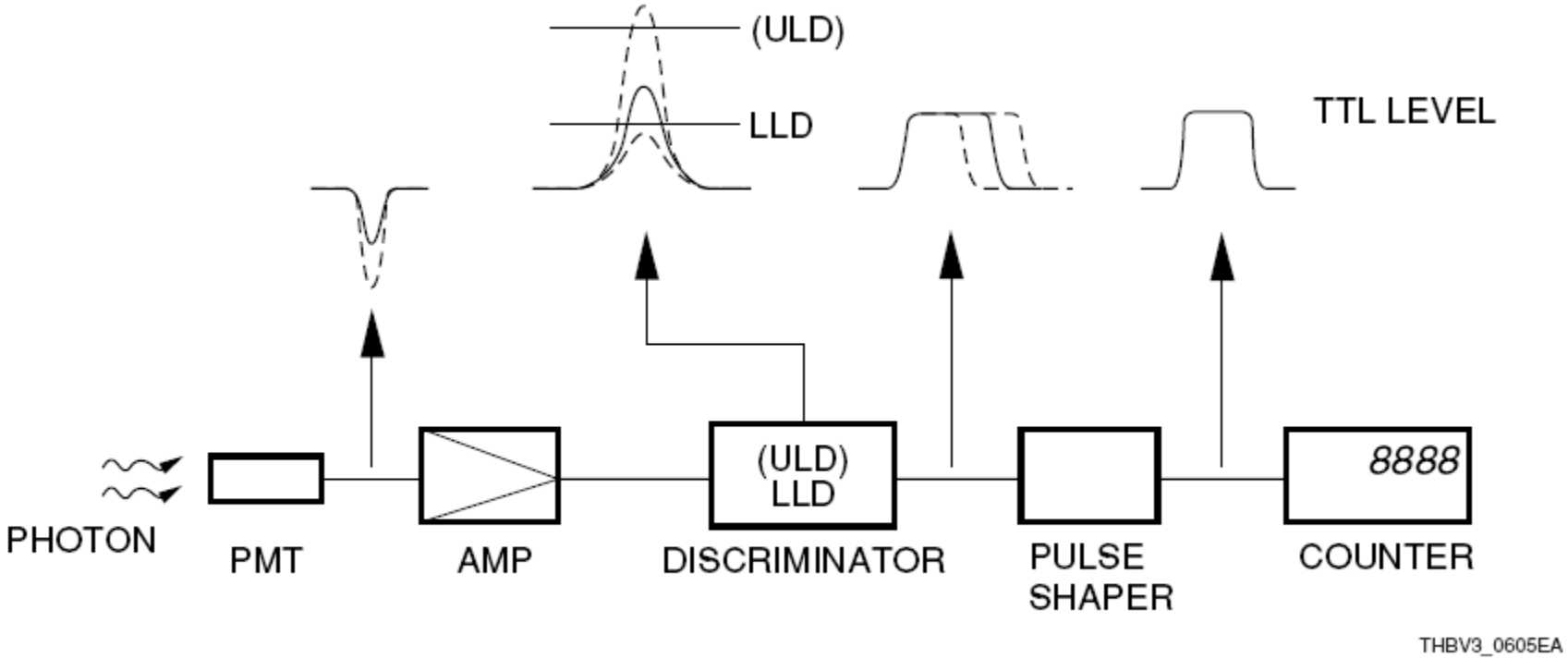


Figure 6-5: Circuit configuration for photon counting

Image Acquisition Using Sensor Arrays

Charge Coupling Devices

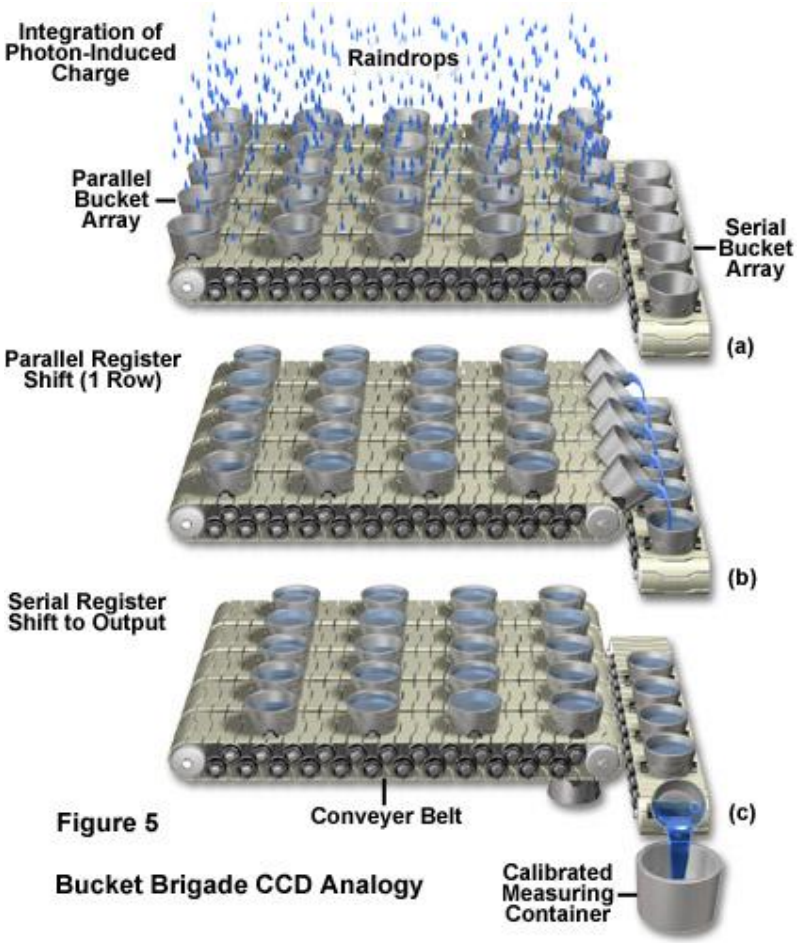


Figure 5

Bucket Brigade CCD Analogy

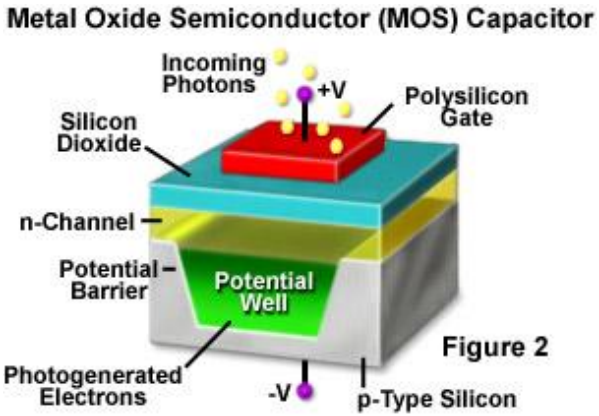


Figure 2

- PN junctions detect photocurrent
- Moving one line forward
- A pixel is read out by the reading unit

http://www.microscopyu.com/articles/digitalimaging/ccdintro.html

CMOS

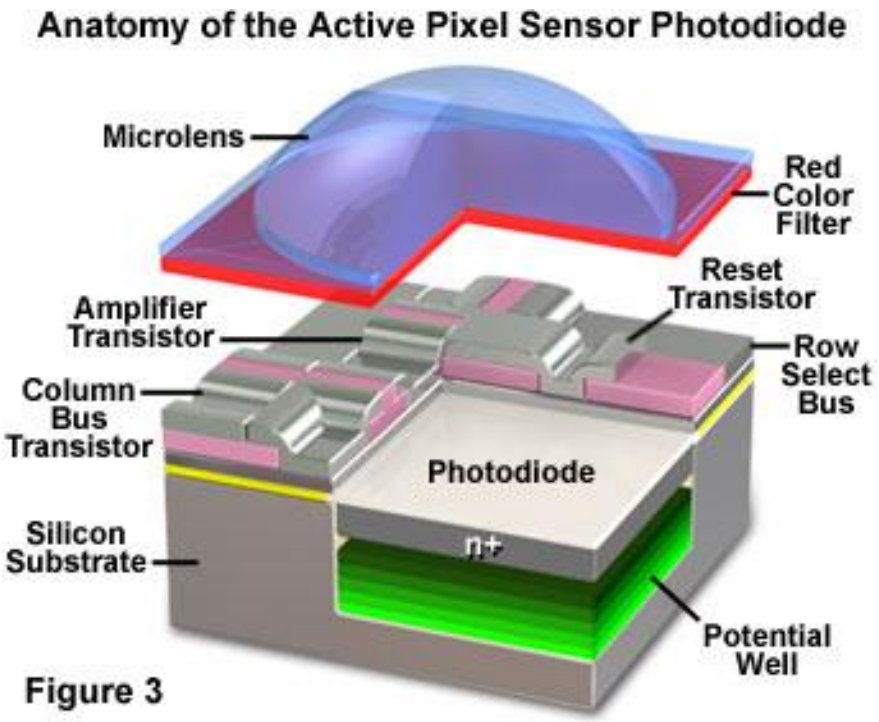
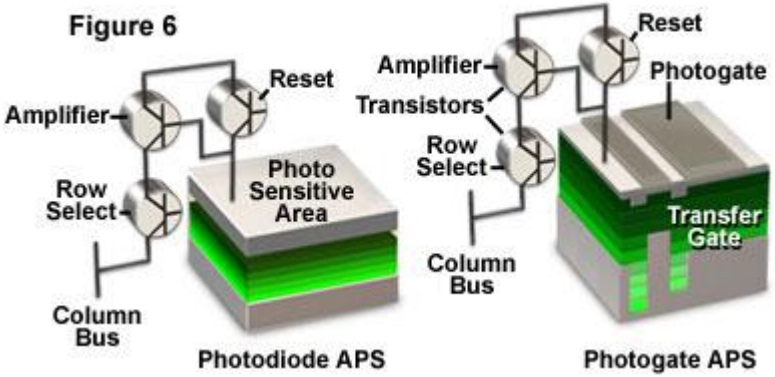


Figure 3

Photodiode and Photogate Structural Features



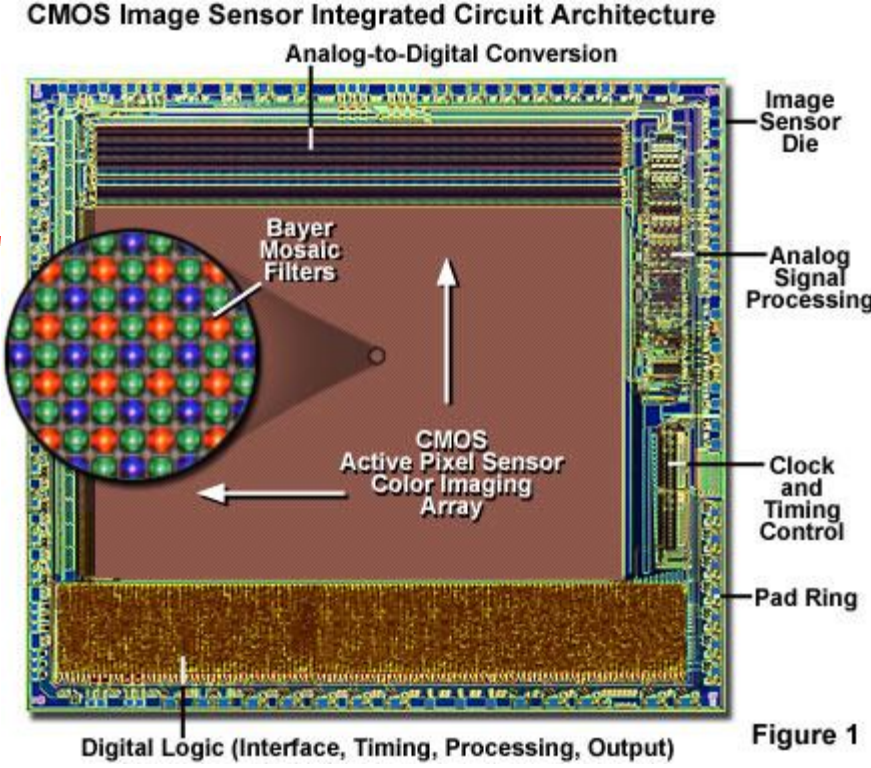
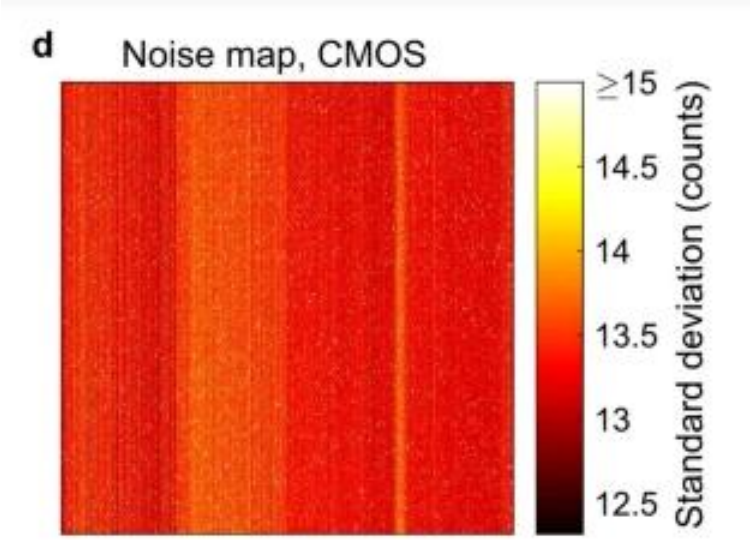


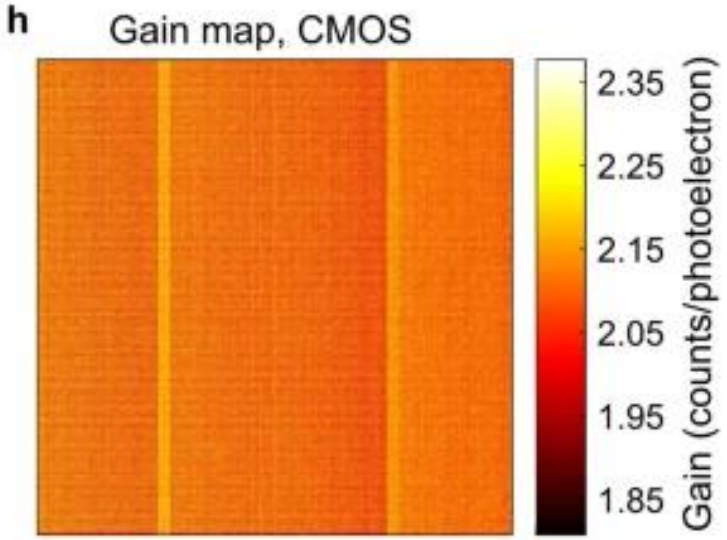
Figure 1

- Low power consumption, low cost, small size
- Small photosensor area
- Un-eveness of the image field

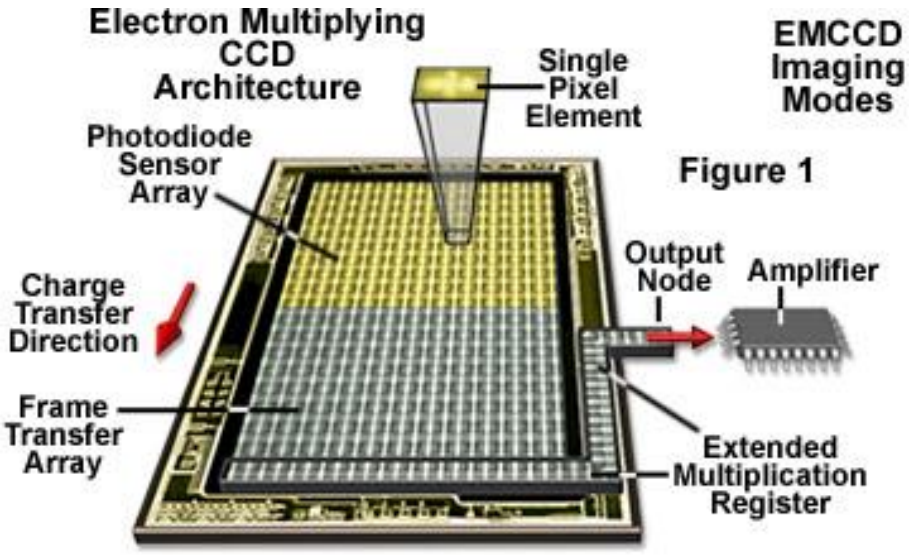
https://www.olympus-lifescience.com/en/microscope-resource/primer/digitalimaging/cmosimagesensors/
http://www.dzsc.com/data/html/2006-4-14/2225.html

CMOS Camera Characteristics





Electron Multiplier CCD (EMCCD)



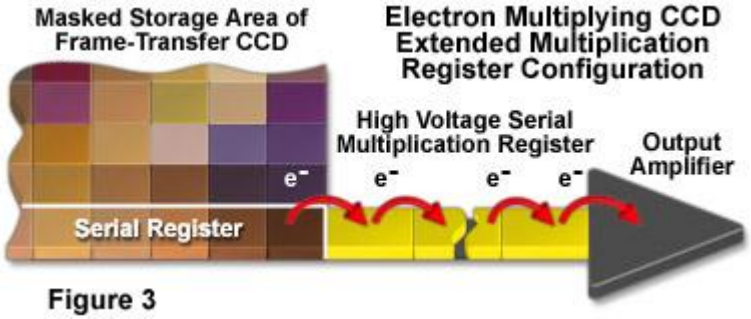


Figure 3

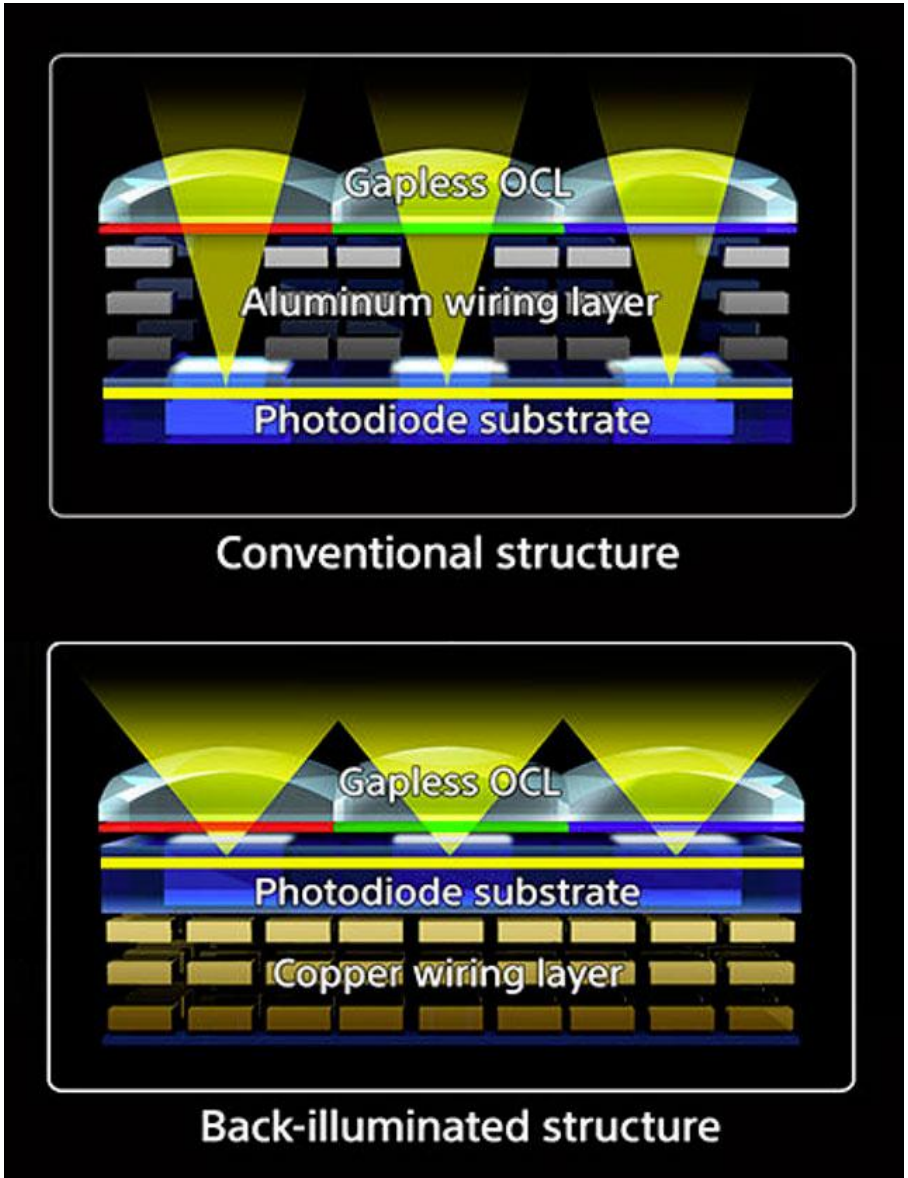
Scientific CMOS

Unlike previous generations of CMOS and CCD-based sensors, sCMOS is uniquely capable of simultaneously offering:

- Extremely low noise
- Rapid frame rates
- Wide dynamic range
- High resolution
- Large field of view

Parameter	sCMOS (Zyla)	Interline CCD	EMCCD
Sensor Format	5.5 megapixel	1.4 to 4 megapixel	0.25 to 1 megapixel
Pixel Size	6.5 µm	6.45 to 7.4 µm	8 to 16 µm
Read Noise	1.2e^- @ 30 frames/sec 1.45e^- @ 100 frames/sec	4 - 10 e^-	< 1 e^- (with EM gain)
Full Frame Rate (max.)	100 frames/sec @ full resolution	3 to 16 frames/sec	~ 30 frames/sec
Quantum Efficiency (max.)	57%	60%	90% 'back-illuminated' 65% virtual phase
Dynamic Range	25,000:1 (@ 30 frames/sec)	~ 3,000:1 (@ 11 frames/sec)	8,500:1 (@ 30 frames/sec with low EM gain)
Multiplicative Noise	None	None	1.41x with EM gain (effectively halves the QE)

Back Illuminated Sensor

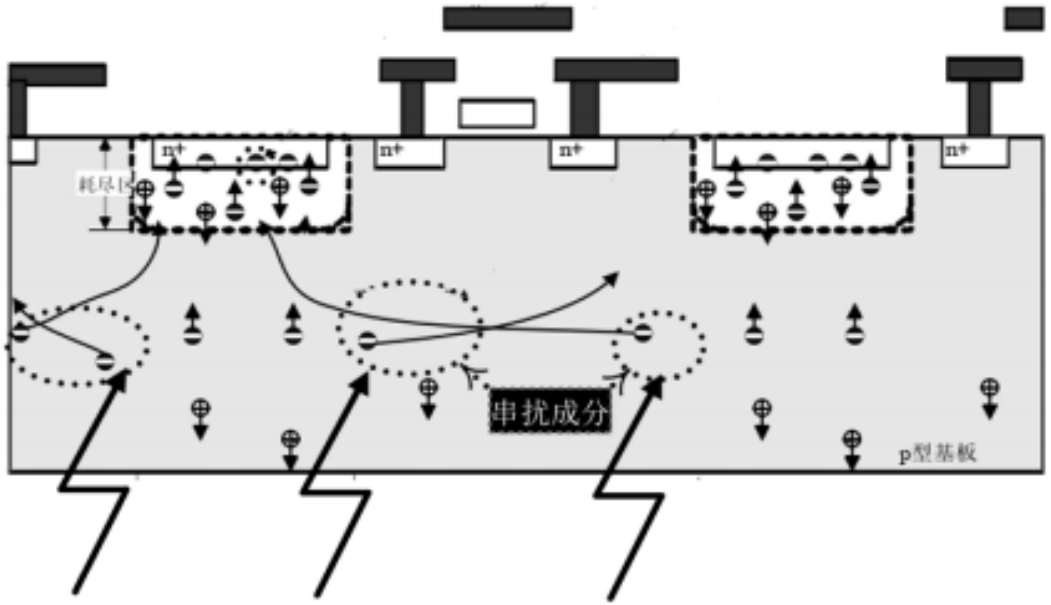


Article | Open access | Published: 15 October 2024

High sensitivity cameras can lower spatial resolution in high-resolution optical microscopy

Henning Ortkrass, Marcel Müller, Anders Kokkvoll Engdahl, Gerhard Holst & Thomas Huser

Nature Communications **15**, Article number: 8886 (2024) | Cite this article



背照式传感器像素中的电学串扰

CCD vs. CMOS vs. EMCCD vs. sCMOS

Architecture explains the choices; specifications depend on the camera.

	CCD	CMOS	EMCCD	sCMOS
Readout	Shift charge to output node(s)	Pixel amplifiers; parallel paths	CCD + EM register before readout	CMOS optimized for measurement
QE	Design & wavelength	Design & wavelength	Often high (back-illuminated)	High in many BSI designs
Read noise	Low at slow readout	Strongly model-dependent	Input-referred noise reduced by gain	Low, including at high speed
Multiplication noise	No EM stage	No EM stage	Yes: stochastic electron gain	No EM stage
Speed	Serial readout can limit rate	Fast parallel readout	Serial readout can limit rate	Fast parallel readout
FOV*	Sensor-format dependent	Wide formats available	Often smaller scientific arrays	Large scientific arrays common
Pixel nonuniformity	PRNU / DSNU still present	Pixel / column variation	CCD variation + gain statistics	Pixel noise / gain need calibration
Microscopy examples	Long-exposure fluorescence	Routine brightfield	Very weak signals; single molecules	Live cells; light-sheet

*Object-space FOV = sensor width / magnification. sCMOS is a CMOS class.

Compare QE, noise, speed and sampling under the same imaging conditions.

Where Does Image Noise Come From?

Physical source first. Statistical model second.

Physical source	What fluctuates?	Useful model
Photon shot noise	Photon arrival / detected count	Poisson
Dark current	Thermally generated electrons	Poisson count
Read noise	Electronic readout baseline	Often Gaussian
Amplification noise	Stochastic EM multiplication	Gamma approximation*

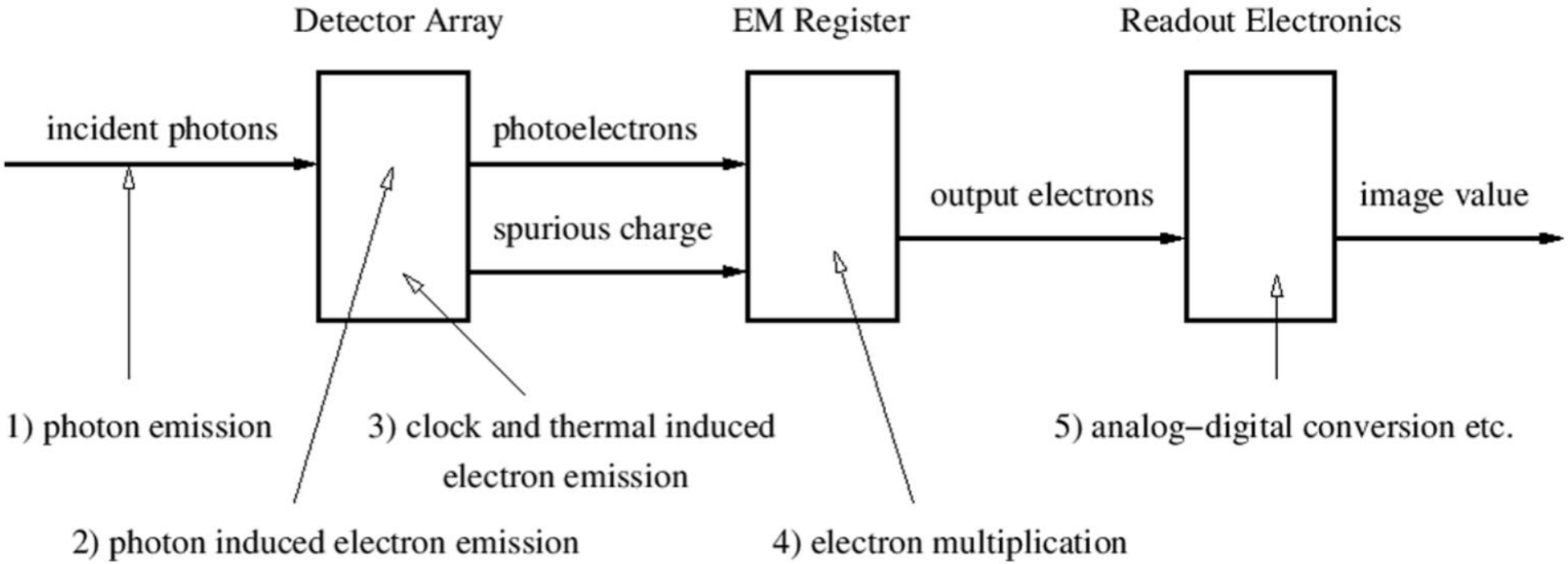
QE describes detection probability

Fixed photon count → Binomial detection; Poisson input → Poisson output.

*EMCCD output conditioned on input electrons at high EM gain; not all amplifiers.

Independent noise variances add: $\sigma^2_{total} = \sigma^2_{shot} + \sigma^2_{dark} + \sigma^2_{read} + \ldots$

Noise Sources



1) Poisson distribution; 2) Binomial distribution; 3) Poisson distribution;
4) Gamma distribution; 5) Gaussian distribution.

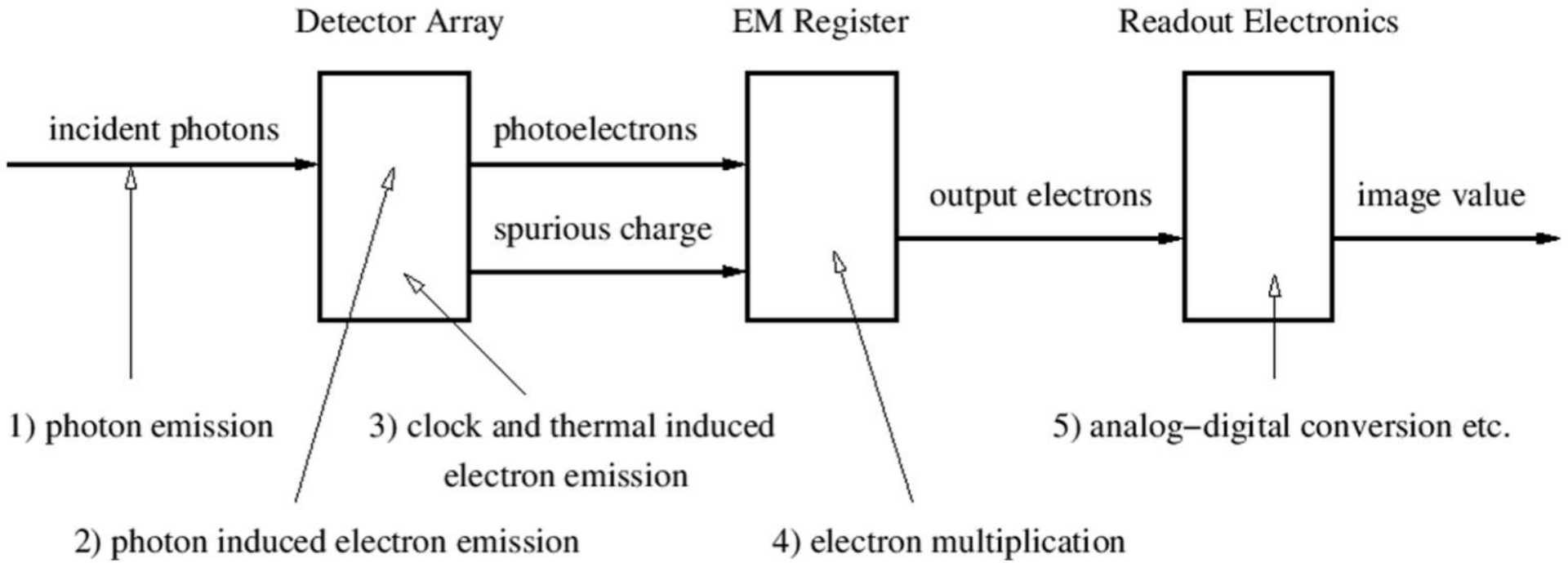
Incident photons n_{ph} :

$$p(n_{ph}; \mathbf{i}) = P(n_{ph}; \lambda = \mathbf{i}).$$

i : expected photon emission

Shot noise

Noise Sources



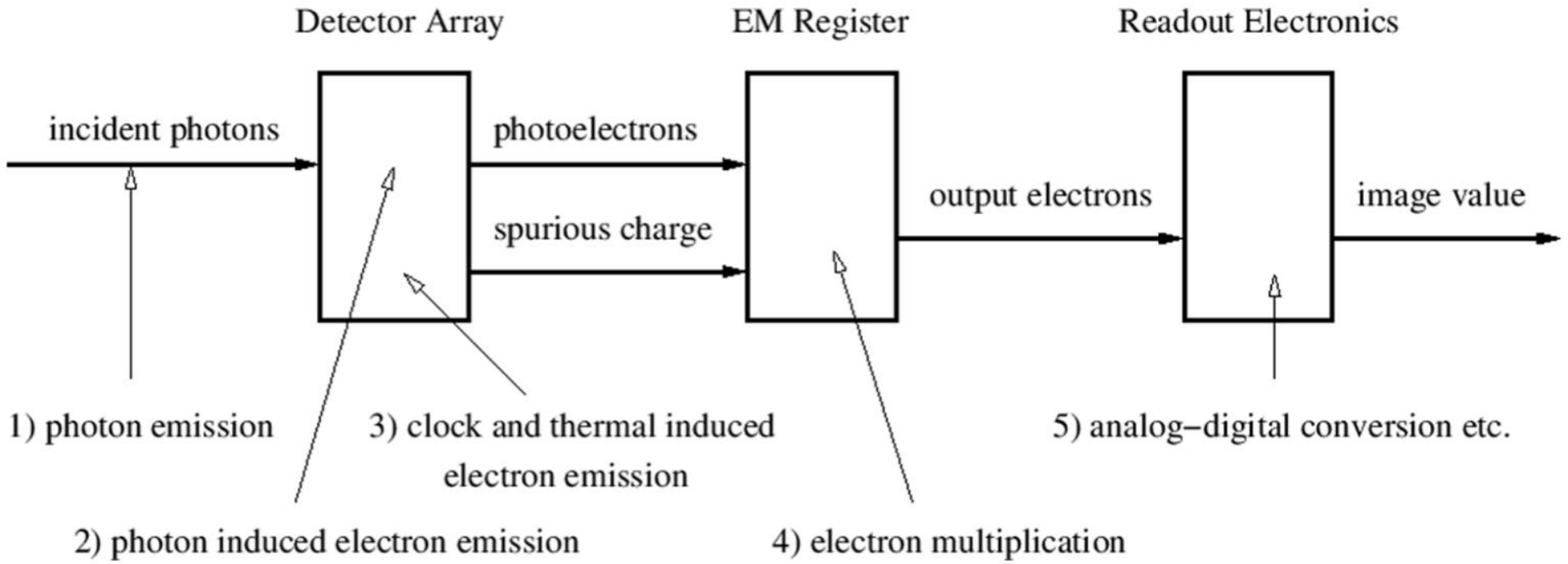
1) Poisson distribution; 2) Binomial distribution; 3) Poisson distribution;
4) Gamma distribution; 5) Gaussian distribution.

Photoelectrons n_{pe} : $$p(n_{pe};\mathbf{i},\mathbf{q}) = \sum_{n_{ph}=0}^{\infty} P(n_{ph};\mathbf{i})B(n_{pe};n_{ph},\mathbf{q}) = P(n_{pe};\mathbf{iq}).$$

q: quantum efficiency of the detector in electrons per photon

Detector quantum efficiency

Noise Sources



1) Poisson distribution; 2) Binomial distribution; 3) Poisson distribution;
4) Gamma distribution; 5) Gaussian distribution.

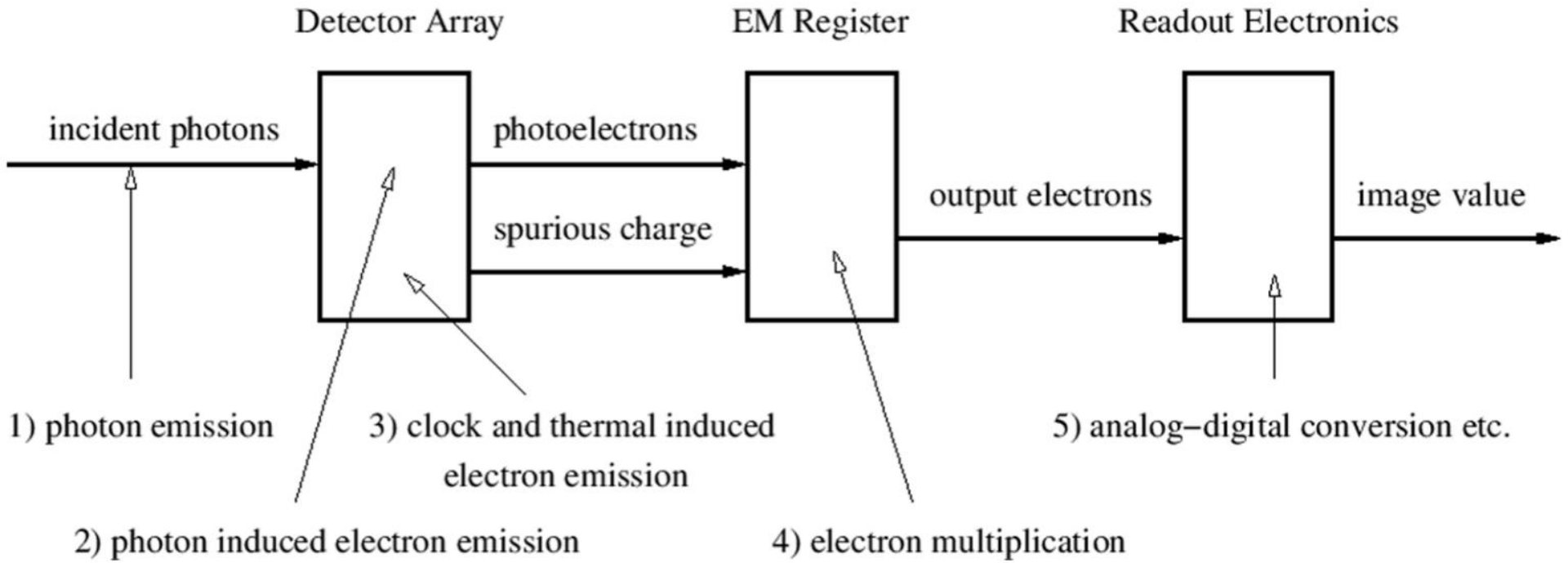
Input electrons before EM register n_{ie} :

$$p(n_{ie};\mathbf{i},\mathbf{q},\mathbf{c}) = \sum_{m=0}^{n_{ie}} P(m;\mathbf{iq})P(n_{ie}-m;\mathbf{c}) = P(n_{ie};\mathbf{iq}+\mathbf{c}).$$

c: spurious charge in electrons

Dark current

Noise Sources



1) Poisson distribution; 2) Binomial distribution; 3) Poisson distribution;
4) Gamma distribution; 5) Gaussian distribution.

Output electrons after EM register n_{oe} :

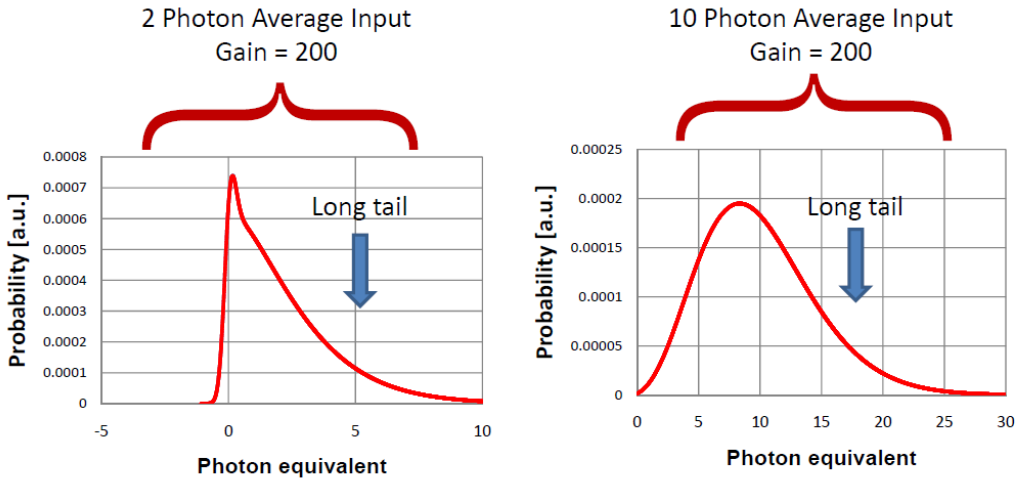
$$p(n_{oe};n_{ie},\mathbf{g}) = \gamma(n_{oe};n_{ie},\mathbf{g}) = {n_{oe}}^{n_{ie}-1}\frac{e^{-n_{oe}/\mathbf{g}}}{\Gamma(n_{ie})\mathbf{g}^{n_{ie}}}.$$

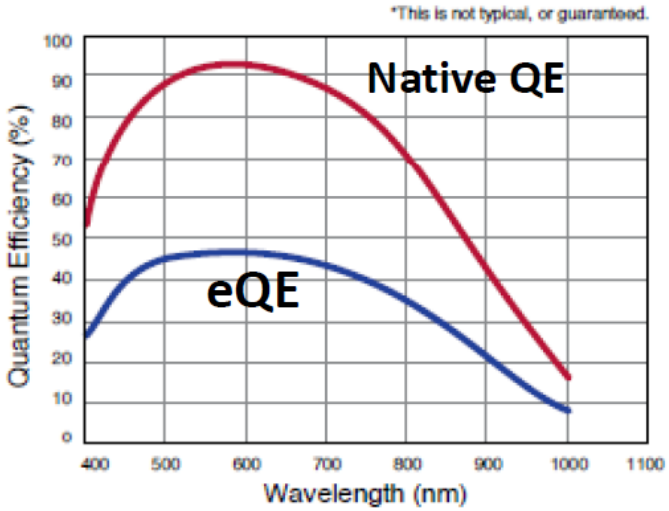
g: EM Gain

Electron multiplication noise

Excess Noise

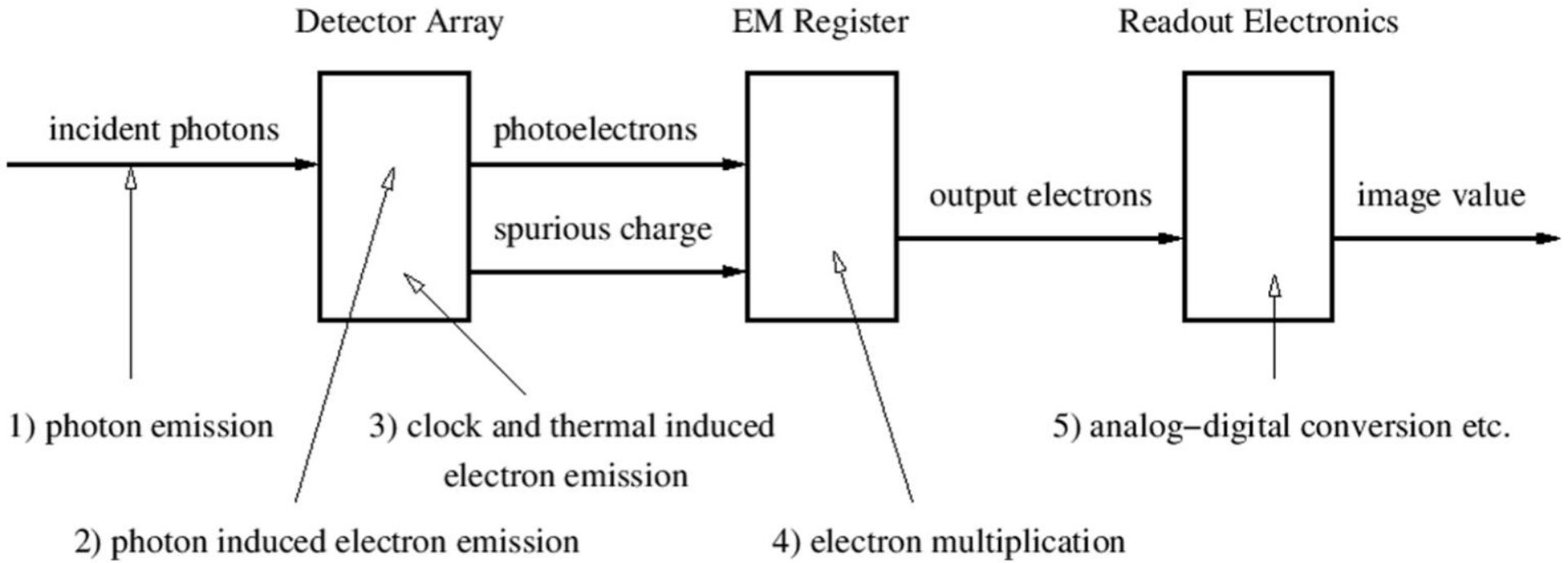
- *Stochastic* EM amplification:
  - Very low noise without input
  - Excess noise effectively doubles photoelectron shot noise ($F_n^2 = 2$)
  - Asymmetric output distribution
    - At low light, peak output is much below mean
    - Long tail





{ Effective QE in EMCCDs }

Noise Sources



1) Poisson distribution; 2) Binomial distribution; 3) Poisson distribution;
4) Gamma distribution; 5) Gaussian distribution.

Image counts: $N(\mathbf{f}n_{ic}; n_{oe}, \mathbf{r})$

Normal distribution

f: A/D factor
r: readout noise in electrons

Readout noise

Simple (Pixel) SNR Equation

$$SNR = \frac{QE * S}{\sqrt{F_n^2 * QE * (S + I_b) + (N_r / M)^2}}$$

Terms included:

QE: Quantum Efficiency

S: Input Signal Photon Number (photon/pixel)

F_n: Noise Factor
(= 1 for CCD/sCMOS and √2 for EM-CCD)

N_r: Readout Noise

M: EM Gain (=1 for CCD / CMOS)

I_b: Background

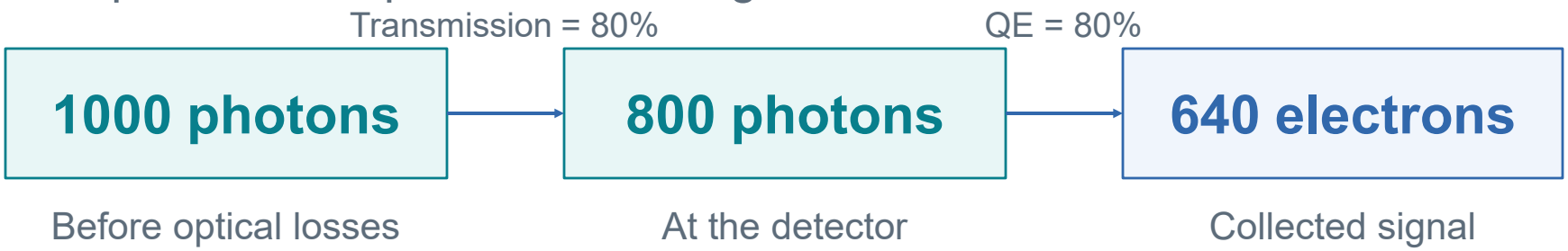
Not included:

Dark Noise: Dark current X time; considered negligible

Photo response non uniformity: necessary for *image* SNR

Photon Budget

One pixel, one exposure — average counts



Photon shot noise $\sqrt{640} \approx 25.3\ e^-$ rms

Read noise (assumed) $3\ e^-$ rms

Total noise $\sqrt{(640 + 3^2)} \approx 25.5\ e^-$ rms

SNR ≈ 640 / 25.5 ≈ 25

ADC: 1 e^-/DN + 100 DN → 740 DN

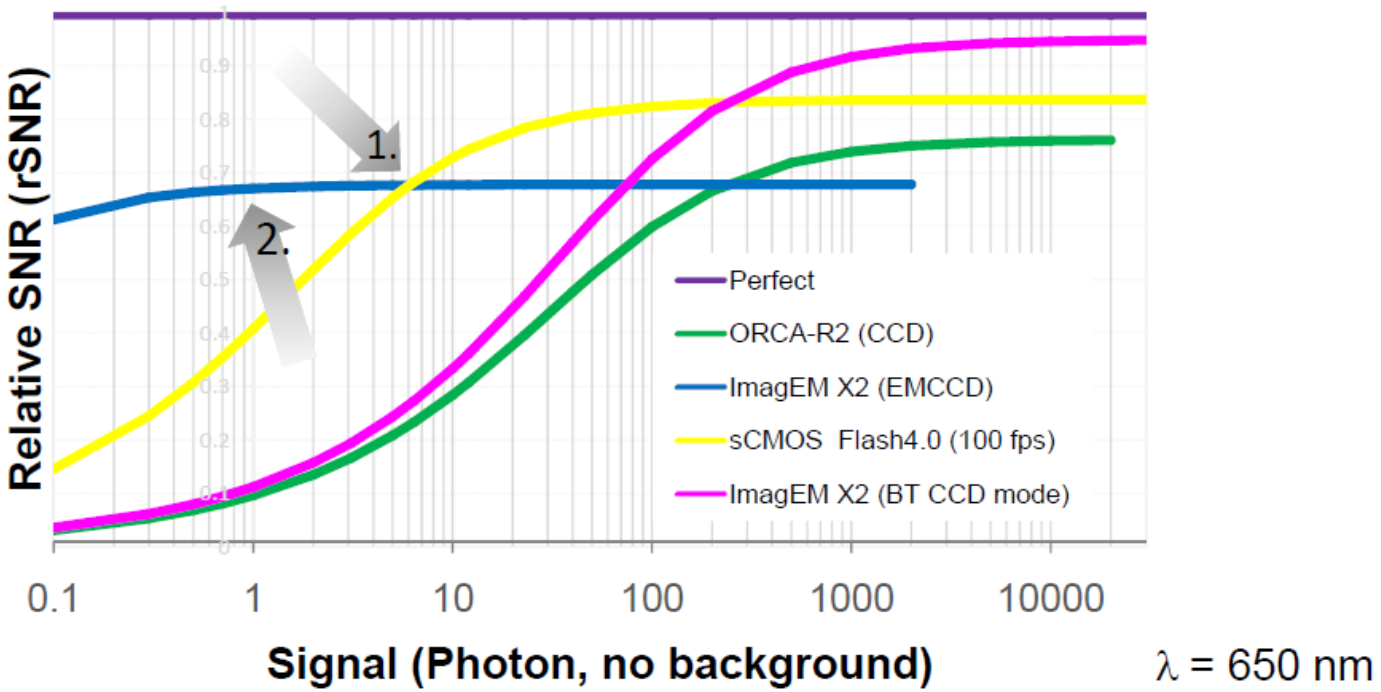
Assume Poisson illumination; negligible background, dark and quantization noise.

Which noise dominates? What if we detect only 4 e^-?

SNR Comparison

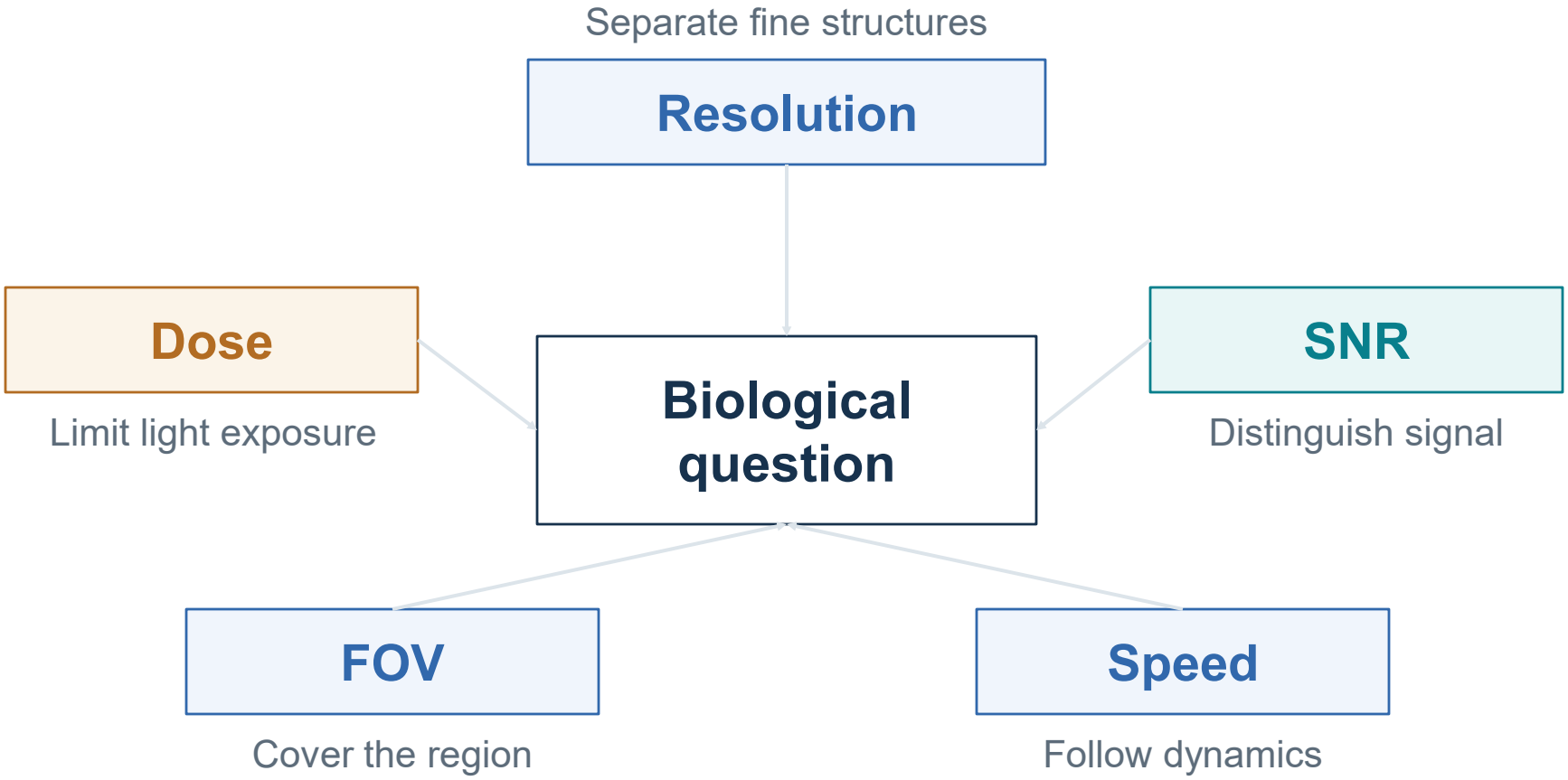
1. A camera with the highest SNR at the lowest light level may not be the best at higher light levels

2. The SNR of an EMCCD above 1 electron/pixel is comparable to a camera with QE_{eff} =QE/2 due to excess noise from EM gain.



λ = 650 nm

What Makes a Good Imaging System?

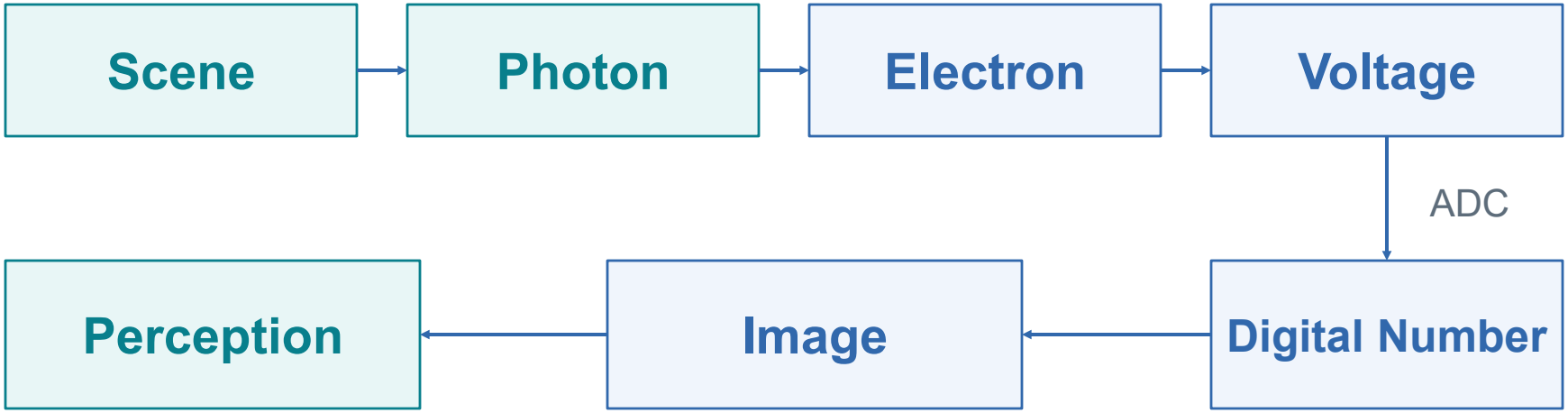


Shorter exposure → fewer photons → lower SNR (fixed flux)

Higher magnification → finer sampling, smaller FOV (same sensor)

There is no perfect imaging system.

Take-home Messages



1 A digital image is a measurement.

2 Signal conversion also introduces uncertainty.

3 Image quality depends on the whole system.

Before interpreting a pixel value, ask how it was acquired.